

Adverse Selection Among Early Adopters and Unraveling Innovation

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Abstract

I provide an equilibrium analysis of “selection markets”: where consumers not only vary in how much they are willing to pay, but also in how much they cost to the seller. I provide comparative statics that order adversely selected markets by their equilibrium prices, quantity, and welfare. I use these results to provide a joint explanation for three empirical phenomena: low uptake of existing products, slow demand for new products, and market inactivity despite unmet demand. I characterize when early adopters are more adversely selected in new markets and show evidence of this in the expansion of the market for deferred income annuities. This adverse selection lowers demand, increases costs, and may lead markets to unravel prematurely. With endogenous market entry for new products (e.g., reverse mortgages, annuities), market power works to ameliorate this cost of adverse selection and extended patents serve as de facto time-varying subsidies.

Keywords: Adverse selection, Insurance, Annuities

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Selection markets are ubiquitous. These are markets where consumers differ in the costs they impose on a firm when they purchase a good, including annuities, insurance, credit markets, financial securities, used goods, and even labor markets.¹

Moreover, these markets are often characterized by low utilization, for example in annuities. Empirical work typically rationalizes this under-utilization through low consumer demand.² However, this explanation overlooks an essential aspect: the supply side's failure to offer more attractive products in response to unmet consumer demand. I show how adverse selection in immature markets exacerbates initial low demand and causes unraveling. Even if a market would be fully efficient if it could reach maturity, higher adverse selection among early adopters prevents it from maturing.

I show that adverse selection is larger among early adopters in young selection markets. In immature markets, potential consumers have less confidence in the product and benefit less from taking up new products. Consequently, demand in immature markets is lower than in mature markets. Crucially, demand falls, but selection patterns also change. Early adopters, those willing to buy the product despite their lower initial private valuations, are also the consumers that impose the highest costs on firms, making adverse selection generically higher in immature markets.

I thus provide a *joint* explanation for three important empirical regularities in selection markets: first, the low take up of existing products, second, sluggish demand when new products are introduced, and, third, market inactivity despite substantial unmet demand for new or modified alternatives. For example, [Zinman \(2014\)](#) highlights missing rungs in the consumer lending ladder, while [Ameriks et al. \(2016\)](#) establish unmet demand for a better formulation of long-term care insurance. Similarly, [Cocco and Lopes \(2015\)](#) establish demand for alternative reverse mortgage products, while

¹[Einav et al. \(2021\)](#) provide an overview.

²See [Lockwood \(2012\)](#) and [Pashchenko \(2013\)](#) for annuities, [Ameriks et al. \(2016\)](#) and [Lockwood \(2018\)](#) in long-term care insurance, and both [Nakajima and Telyukova \(2017\)](#) and [Cocco and Lopes \(2020\)](#) in reverse mortgage markets. Alternatively see [Einav et al. \(2010\)](#) for results emphasizing adverse selection.

Michaud and Amour (2023) document potential demand for risk management products that bundle existing insurance. Yet, Nakajima and Telyukova (2017) and Webb (2011) document very low take up rates for novel reverse mortgage and annuity products, respectively. Similarly, despite being sold to the public since 1954, the market for variable annuities (bundling tax-favorable saving, insurance, and indexed annuity options) remained small until the early 1990s (Brown and Poterba 2006).³

This paper's first two contributions are technical and provide the building blocks to understand the important empirical phenomena outlined above. I first provide comparative statics results across selection markets with the same underlying distribution of cost-relevant types using their joint distributions of model primitives. This allows me to develop an ordering over adversely selected markets in terms of their equilibrium quantity, prices, selection, and welfare. Moreover, I show this in an environment that admits a graphical representation and I provide sufficient conditions for a graphical analysis of the selection problem with multi-dimensional heterogeneity. These are two key contributions of the paper.

I provide explicit and economically interpretable sufficient conditions for non-parametric families of joint distributions which characterize shifts in the demand and cost curves in selection markets. These conditions additionally provide sufficient conditions for an ordering over equilibrium adverse selection, demand and welfare between markets. In adversely selected markets under-insurance rises and consumer surplus falls when sorting on cost increases or private valuations decrease. I apply these comparative static results to characterize differences between mature and immature markets, nevertheless the insights apply equally to markets indexed by space and time or for different products.

³More generally, concern about missing innovation in insurance markets dates back over half a century: Rudelius and Wood (1970) identify heterogeneous adoption of life insurance innovations, Dorfman (1971) provides evidence of research activity and barriers to innovation, and Murray (1976) surveys insurance companies to identify innovations.

I establish these results in a standard model of insurance demand with multidimensional heterogeneity. While I focus on insurance markets to provide a concrete example, results throughout the paper apply to selection markets in general. In this framework, consumers are heterogeneous along multiple dimensions including costs and risk premia, which determine the surplus from acquiring insurance, as well as other heterogeneity which captures frictions or departures from rationality. I consider adverse selection settings in which consumers' willingness-to-pay for insurance is positively correlated with their costs to insurers.⁴

To study unraveling and entry in novel markets, I focus on mature and immature markets. Mature markets are those that have existed for a long time and are well understood by consumers, while immature markets are new and unfamiliar. The standard framework can be extended to capture the differences between mature and immature markets by allowing consumers' willingness-to-pay for the product to differ across market maturity.

These differences in willingness-to-pay capture objective differences, such as short-term counterparty risk (Briggs et al. 2023) or initial self-insurance patterns that consumers' cannot quickly adjust, which lower the value of purchasing insurance. However, they can also capture subjective differences including consumers' initial perceptions or distrust of new products, for example inertia (see Handel 2013, who studies health insurance) or learning dynamics (see Israel 2005, for the case of learning in automobile insurance). These differences between mature and immature markets lower demand and increase consumer sorting based on costs. This increases adverse selection in immature markets and generates under-insurance in equilibrium with lower adoption and higher costs.

A key insight delivered by this approach is that even would-be efficient markets can unravel during their immaturity and fail to mature. This is a new, dynamic source of

⁴Analogous results can be obtained for advantageous selection.

market failure building on the classic insight of [Akerlof \(1970\)](#). I frame these results in the context of market maturity and use them to compare mature and immature selection markets. These comparative statics show that, under reasonable assumptions on the primitives of immature markets, demand is lower and the cost to serving the market increases for insurers compared with mature markets. This rationalizes sluggish demand for new products within a framework that can also predict the low utilization of existing products and the complete unraveling of these markets.

Given the implications of the stylized model, I combine market data on annuity demand and micro-data on insurance quotes to show that trends in the market for deferred income annuities are in line with the model predictions. In equilibrium, adverse selection falls in a growing, or maturing, market. This is consistent with [Warshawsky \(1988\)](#) who documents declining adverse selection costs for annuities sold to women over the 20th century. Thus, I show that the model predictions are empirically relevant. Subsequently, I build on the theoretical insights to study endogenous market entry in the context of novel, but segmented, selection markets.

I show that unraveling in immature markets gives rise to a free-rider problem – leading to permanently dormant markets. Even though firms have dynamic incentives to enter profitable mature markets, firms may be unwilling to bear the cost of familiarizing consumers with new products in adversely selected markets because other firms compete away profits once it reaches maturity. This happens even though firms know lower willingness-to-pay and increased adverse selection in immature markets is transitory. These results explain the empirical phenomena of missing selection markets even when evidence suggests adverse selection may be limited.

A novel contribution of this paper is to challenge the common understanding of the interplay between market power and adverse selection, a relationship typically analyzed within static frameworks. Traditional static analyses, for example [Starc \(2014\)](#) and [Mahoney and Weyl \(2017\)](#), highlight that adverse selection can mitigate the negative

impacts of market power by constraining firms' ability to extract rents. Nevertheless market power exacerbates under-supply. The simple dynamic analysis, however, reveals a mechanism that works in the opposite direction: market power can, in fact, alleviate dynamic inefficiencies induced by adverse selection. Thus, in this dynamic setting market power becomes a tool to constrain the social cost of adverse selection.

Finally, having identified this novel source of market failure, I consider the role of government intervention in promoting entry and innovation within absent selection markets. I discuss how increasing patent lengths or burdensome regulation can serve as an effective, time-varying subsidy with minimal informational requirements. This solution addresses two fundamental challenges: the inherent absence of data in missing markets and the potential distortions from intervention in mature markets.

Related Literature

The comparative static tools I develop provide non-parametric conditions for an ordering over partial unraveling in Akerlof markets and are used to consider the dynamic implications of firm entry decisions in novel selection markets. [Akerlof \(1970\)](#) studies the failure of price mechanisms and market unraveling in selection markets. An extended no-trade theorem for [Rothschild and Stiglitz \(1976\)](#) markets is provided by [Hendren \(2013\)](#), while [Attar et al. \(2021\)](#) provide necessary and sufficient conditions for no-trade theorems in adverse selection markets.⁵ [Kong et al. \(2023\)](#) show how adverse selection contributes to non-entry among differentiated firms, even absent fixed costs. This paper extends non-entry results to cases where adverse selection declines over time.

I study changes in the distributions of choice frictions similarly to [Handel et al. \(2019\)](#). I provide comparative statics for non-local changes in both the distribution of choice

⁵A sizable literature studies endogenous contract terms under adverse selection (e.g. [Rothschild and Stiglitz 1976](#); [Handel et al. 2015](#); [Veiga and Weyl 2016](#); [Azevedo and Gottlieb 2017](#)) or screening with both intensive and extensive margins of choice ([Geruso et al. 2023](#)). This is not the focus of this paper.

frictions and the insurance value of contracts across markets. I then use this approach to build a partial ordering of equilibrium prices, quantities and welfare in terms of model primitives. [Handel et al.](#), however, study local reforms (including information interventions) that alter consumer's choice frictions and provide sufficient statistics formulas for their welfare impacts. As in [Handel et al.](#) and [Spinnewijn \(2017\)](#) I show how changes to model primitives lead to the rotations and shifts to cost and demand that are studied in other contexts (e.g. [Starc 2014](#); [Mahoney and Weyl 2017](#)).

I show that a novel free-rider problem manifests in the dynamics of adversely selected markets. Importantly, this reverses standard intuition for the interaction of market power and adverse selection. At any point in time, adverse selection impacts firm's ability to exploit market power as discussed by [Starc \(2014\)](#) and [Mahoney and Weyl \(2017\)](#). Moreover, market power exacerbates the under-supply driven by adverse selection. Here, however, the presence of market power early in a product's life cycle can incentivize entry when would-be efficient markets fail to mature.

I examine the implications of changing adverse selection over time and study policies promoting initial market entry. These results emphasize that inter-temporal cross-subsidization can be optimal under missing markets. These policies are related to, but differ from, those designed to mitigate selection ([Geruso and Layton 2017](#), review these policies in health insurance markets).

1. A Model of Equilibrium in the Insurance Market

I consider an [Akerlof \(1970\)](#) model of insurance in the style of [Einav et al. \(2010\)](#) and [Einav and Finkelstein \(2011\)](#). A single perfectly competitive and risk-neutral insurer sells insurance to a unit continuum of heterogeneous individuals at a common price p . The terms of the insurance contract are exogenous and common across all individuals.

Demand. Individuals are heterogeneous along multiple dimensions. Namely, their risk type π_i which determines the expected cost to the insurer, their risk premium r_i , and potential demand frictions ε_i . These are private information for individual i .⁶ An individual buys insurance if and only if their *subjective* willingness-to-pay, defined as

$$\omega_i \equiv u(\pi_i, r_i, \varepsilon_i) = u(\underbrace{\pi_i + r_i + \varepsilon_i}_{\text{insurance value}}), \quad (1)$$

for any strictly increasing utility function $u(\cdot)$, exceeds the price of the insurance. In addition to their risk type, individual i 's risk premium, r_i , determines the surplus in the insurance market.

Following [Spinnewijn \(2017\)](#), individual choices depend on additively separable demand frictions, ε_i , which can distort insurance demand. This approach is agnostic about the nature of demand frictions, nesting a number of important model features including borrowing constraints, but also inertia, subjective beliefs, and bounded rationality. These frictions are heterogeneous and can lead individuals to overvalue ($\varepsilon > 0$) or undervalue ($\varepsilon < 0$) the insurance contract. As risk premia and frictions have identical effects on consumer choices I focus on consumers' perceived risk premia, net of any behavioral influences, given by $\tilde{r}_i \equiv r_i + \varepsilon_i$, for the majority of the analysis.

Thus, demand is given by

$$D(p) = 1 - F_\omega(p) = \mathbf{1}[\omega > p], \quad (2)$$

where F_x denotes the CDF of variable x .⁷

⁶Conditioning on observable characteristics correlated with π_i is not always legal. Nevertheless, this can be interpreted as a model of insurance contracts in each partition of the observable characteristic space.

⁷Similarly, μ_x and σ_x^2 will denote the mean and variance of x 's distribution.

Supply. The cost of providing insurance at price p is determined by the risk types of individuals purchasing insurance at that price.⁸ Consequently, *average* and *marginal* costs are given by:

$$AC(p) = E(\pi | \omega \geq p) \quad (3)$$

$$MC(p) = E(\pi | \omega = p) . \quad (4)$$

When willingness-to-pay, ω , is increasing in the cost to the insurer, π , there is adverse selection; marginal costs increase in price and the MC curve is downward sloping.

With multidimensional heterogeneity, the critical assumption for adverse selection is this dependence between individual willingness-to-pay and costs (See Fang and Wu 2018). In general, this depends on the shape of the copula which represents their joint distribution.⁹ Copulas encode the correlational structure between π_i , r_i , and ε_i without imposing a specific parametric family of distributions and decouple assumptions on dependence properties from those on marginal distributions.¹⁰ They encode this structure as restrictions on the shape of rank dependence or sorting in quantile space (Anderson and Smith 2024). Using copulas allows for the role of *sorting* to be disentangled from the distribution of costs and risk premia.

The introduction of copulas is more than a technical abstraction. Although results

⁸For simplicity, I abstract from administrative loadings, or fixed costs which don't affect the main result.

⁹Copulas have many uses in economics and finance – particularly in the context of selection and dependent risks. Recent examples include competing hazard models (Kim 2021), and multi-dimensional skills (Gola 2021). The main representation result is Sklar's theorem, which I reproduce below.

Sklar's Theorem: Let F be a joint distribution function for a d -dimensional random vector $(X_1, X_2, \dots, X_d)$, and let F_i be the marginal distribution function for the i -th component X_i . Then, there exists a copula C such that for any $x_1, x_2, \dots, x_d$ in the real line:

$$F(x_1, x_2, \dots, x_d) = C(F_1(x_1), F_2(x_2), \dots, F_d(x_d))$$

Moreover, if X is continuous, then the Copula C is unique.

¹⁰Copulas are invariant to rank-preserving transformations. Therefore, results immediately apply to multiplicatively separable utility, $u = \exp(\cdot)$, and when insurer costs are transformed cost types.

can be characterized for the special case of joint-normality, this comes at the cost of restricting the skewed risks that consumers experience and insurers insure. Moreover, normality rules out heavy-tails in the distribution of risks. These heavy tails are an important concern in actuarial modeling because tail events have large impacts on the balance sheets of insurers and financial institutions despite their infrequency. Thus, although the language of copulas increases the complexity of the results it also allows the general framework to describe economically and empirically relevant settings. To maximise accessibility, I present the proofs for the special case of joint-normality in the main text.

Equilibrium. To simplify notation, I assume demand and cost curves are continuous and monotonic which implies the equilibrium is unique if it exists. The equilibrium price is given by the break even price that pools expected costs

$$p^* = p : D(p) = AC(p) . \quad (5)$$

Consequently, the equilibrium allocation may differ from an efficient allocation. An efficient allocation insures everyone whose willingness-to-pay exceeds their expected cost ($\omega_i > \pi_i$), those for whom the demand curve lies above the marginal cost curve as shown in Panel (A) of Figure 1.

Furthermore, a positive equilibrium price does not necessarily exist. The insurance market unravels (Akerlof 1970) when individuals are not willing to purchase insurance at the pooled cost of those with higher demand. This occurs when the demand curve lies entirely below the average cost curve (Hendren 2013) as shown in Panel (B) of Figure 1. I now provide sufficient conditions for monotonic cost curves and adverse selection under arbitrary marginals – a prerequisite for unraveling. In subsequent sections I show how characteristics of immature and mature markets affect this unraveling.

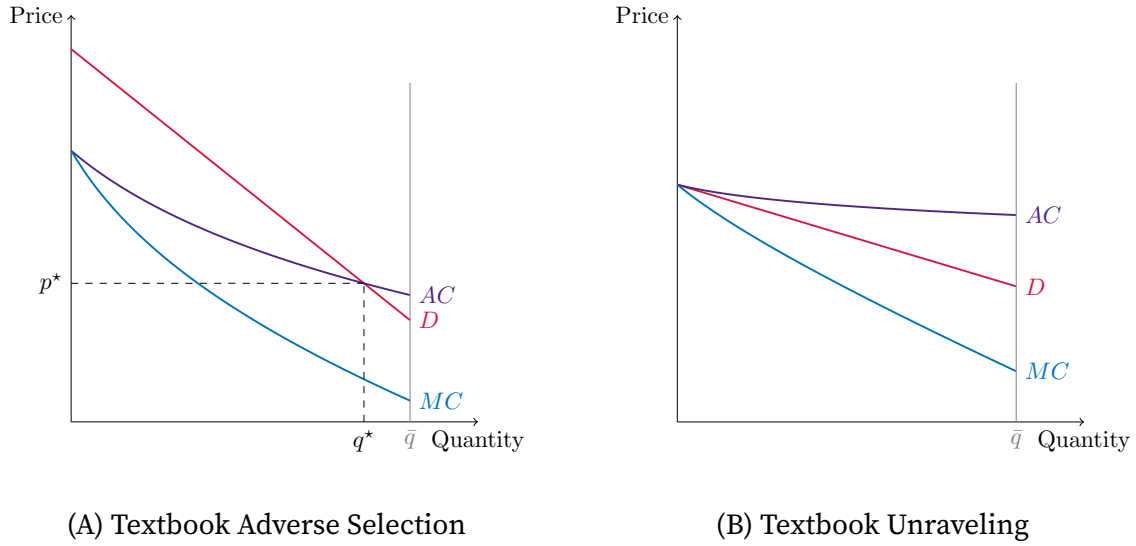


FIGURE 1. Equilibrium in the Insurance Market Without Demand Frictions

1.1. Sufficient Conditions for Monotonicity and Adverse Selection

As Fang and Wu (2018) note, selection problems with multidimensional types do not generically satisfy monotonicity.¹¹ Despite this, a large body of empirical work following Einav et al. (2010) and Einav and Finkelstein (2011) rely on price variation to estimate marginal cost curves (Einav and Finkelstein 2023 provide a recent review of the literature adopting this framework). These studies adopt a graphical approach and use the marginal cost curve as a sufficient statistic for consumer heterogeneity – an appealing insight that only holds under monotonicity. The following proposition provides sufficient conditions on the joint-distributions for the validity of this approach without placing any restrictions on the marginal distributions. Alternatively, these can be viewed as the precise assumptions for imposing shape restrictions in non-parametric estimation.

¹¹This motivates Fang and Wu to consider local adverse (and advantageous) selection in a multidimensional context. Their Definition 2 corresponds to condition (i) in Proposition 1. The global conditions outlined in Proposition 1 have a natural analogue for monotone regions and local adverse selection. The conditions below are necessary and sufficient conditions for their requirements on the conditional densities.

PROPOSITION 1 (Adversely Selected Markets). *Let copula $\hat{C}$ denote the dependence between cost type, π_i , and willingness-to-pay, ω_i . Then*

- i. *[Component-wise Concavity] The market is adversely selected, $MC(p)$ is increasing in price, if $\hat{C}(u, v)$ is concave in v for all $u \in (0, 1)$.*
- ii. *[Corner Set Monotonicity] Average costs, $AC(p)$, are monotonically increasing in price if $(u - \hat{C}(u, 1-v))/v$ is increasing in v for all $u \in (0, 1)$.*

The formal proof is given in Appendix A. The following simple example illustrates this for the joint normal distribution. Suppose π and $\tilde{r}$ are multivariate normal with correlation $\rho_{\pi, \tilde{r}}$. Then the joint (normal) distribution of cost type and willingness-to-pay is given by,

$$\begin{pmatrix} \pi_i \\ \omega_i \end{pmatrix} \sim N \left[\begin{pmatrix} \mu_\pi \\ \mu_\omega \end{pmatrix}, \begin{pmatrix} \sigma_\pi^2 & \rho \sigma_\pi \sigma_\omega \\ \rho \sigma_\pi \sigma_\omega & \sigma_\omega^2 \end{pmatrix} \right],$$

where

$$\rho = \frac{\sigma_\pi + \rho_{\pi, \tilde{r}} \sigma_{\tilde{r}}}{\sigma_\omega} > 0. \quad (6)$$

Consequently, the marginal cost curve has the following closed form expression:

$$MC(p) = E(\pi | \omega = p) = \mu_\pi + \rho \frac{\sigma_\pi}{\sigma_\omega} (p - \mu_\omega), \quad (7)$$

which is increasing in p when $\rho_{\pi, \tilde{r}}$ (consequently ρ) is positive. In this case, average costs are the expected value of a truncated normal distribution. The average cost curve

is

$$AC(p) = E(\pi | \omega \geq p) = \mu_\pi + \rho \sigma_\pi \frac{\phi(\frac{p - \mu_\omega}{\sigma_\omega})}{1 - \Phi(\frac{p - \mu_\omega}{\sigma_\omega})}, \quad (8)$$

which again increase in p for positive $\rho_{\pi, \tilde{r}}$.

More formally, conditions (i) and (ii) restrict the (scale invariant) rank dependence *within* markets and formalize the intuitive notion that larger values of X tend to go with larger values of Y . It implies positive quadrant dependence and, therefore, both positive values of standard correlation coefficients (e.g. Pearson, Kendall's τ , and Spearman's ρ) and regression coefficients.

For parametric copula families these conditions restrict the parameter space, for example requiring the correlation coefficient is positive for the normal copula. This relates to the intuitive misconception that positive correlation is sufficient for adverse selection, however, this is in fact a consequence of the conditions in Proposition 1.

While these conditions are easily verifiable in any empirical setting or application, they are somewhat unintuitive. The following lemma provide stronger sufficient conditions relying on a commonly invoked assumption in economic applications. Furthermore, Appendix A shows how the joint distribution of ω and π can be derived from assumptions on the joint distribution of π and $\tilde{r}$.

Lemma 1 provides a sufficient condition for the first two properties using the familiar monotone likelihood ratio property (Milgrom 1981).¹²

LEMMA 1 (MLRP). *All joint distributions satisfying the Monotone Likelihood Ratio Property, that is for any $\omega_1 \geq \omega_2$ and $\pi_1 \geq \pi_2$, all densities satisfying $f(\omega_1|\pi_1)/f(\omega_1|\pi_2) \geq f(\omega_2|\pi_1)/f(\omega_2|\pi_2)$ satisfy conditions (i) and (ii) in Proposition 1.*

Intuitively, the MLRP property allows the insurer to infer higher costs the signal of

¹²A weaker necessary condition for the second property is a log-concave survival function – a common assumption when studying incomplete information (for instance Milgrom and Weber 1982, Chade and Schlee 2012 and Chade and Swinkels 2021).

higher willingness-to-pay. Corollary A1 in Appendix A provides a reinterpretation of the MLRP conditions in terms of the joint distribution of π and $\tilde{r}$.¹³

Finally, it is important to emphasize that this general representation does not assume that choice frictions are independent of other characteristics. Consequently, choice frictions are a reduced form representation of a richer economic model compatible with a range of behavioral microfoundations where frictions are not independent. This includes rational inattention models of endogenous information acquisition and settings in which agents make correlated mistakes affecting both risk premia and insurance decisions (Leive et al. 2022).

2. Comparative Statics Between Markets and Unraveling in Immature Markets

I now provide assumptions on the primitives of insurance markets captured by the joint distribution of π_i and $\tilde{r}_i$. Throughout I assume that the assumptions in Proposition 3 are satisfied. These assumptions allow me to draw precise predictions about supply and demand as well as equilibrium prices, quantities and consumer surplus. The additional assumptions below are sufficient to provide an ordering over these equilibrium outcomes for markets with common distributions of π_i . Thus, while I motivate the assumptions in the context of studying market maturity, the conditions they impose are more widely applicable. They represent assumptions on pairs of markets, such that they can be ordered by equilibrium quantity, price, adverse selection and welfare.

Below, I discuss possible microfoundations under which these primitives may arise endogenously in the context of market maturity and how they capture empirically

¹³This shows that the MLRP condition depends on the copula density function (given by the cross partial derivative of the copula C). The examples that follow shows that because dependence is measured in ranks, the cross partial derivative captures more than the correlation between π and $\tilde{r}$. It additionally encodes information about how jointly heavy-tailed π and $\tilde{r}$ are. When the copula and marginals are gaussian the correlation coefficient is a sufficient statistic to summarize the cross-partial derivative.

relevant features of the individual's and insurer's problems. In a strict sense, these assumptions capture more than just the passage of time. They also capture the idea that the good is potentially viable in its maturity, which is in line with the focus of this paper. However, again it is important to emphasize that these comparative statics extend beyond the specific motivating example of mature and immature markets and are equally valid when markets are indexed by time or space. Equally, similar assumptions allow for the ranking of advantageously selected markets.

2.1. Differences between Mature and Immature Markets

I use superscripts MM to denote the mature insurance market and IM to denote the immature insurance market for a new form of innovative insurance coverage.

The key assumption I make is that willingness-to-pay is depressed in immature markets relative to mature markets – lowering demand. This leads to more adversely selected early adopters and generates additional unraveling in immature markets. This relative lack of demand can come from objective differences in risk premia, r , or differences in demand frictions, ε . I formalize this in Assumption (1) which defines cases where the marginal distribution of risk premia net of frictions are ordered in a stochastic sense, where $X \preceq Y$ denotes the relation Y first order stochastic dominates X .

ASSUMPTION 1. *The perceived risk premia (net of any behavioral influences) of insurance contracts are lower for immature markets. $\tilde{r}^{MM} \succeq \tilde{r}^{IM}$*

These differences can stem from either differences in the objective risk premia, r , or demand frictions, ε , and I begin by motivating differences in the first component of net premia, $\tilde{r}$. Three channels generate objective differences between the risk premia for immature and mature markets. First, individual portfolios may not be optimized to take advantage of an insurance product when it is first introduced. For example, an

individual may accumulate savings to self-insure in the absence of formal insurance. Thus, the benefits to insurance purchase are larger when other choices can adjust to fully capitalize on the product.

Second, there may be objective uncertainty about the probability that all claims will be covered, the hassle costs associated with claiming, or the likelihood of reclassification risk. For example, in health insurance markets, there is uncertainty over, and heterogeneity in, the coverage of new drugs or procedures. This uncertainty is likely larger for new types of insurance and lowers the risk premia.

Third, objective counterparty risk may be larger in immature markets, relative to mature markets where insurers typically trade with repeated cohorts of potential insurees. In an immature market insurers receive liquid assets now (in the form of premia), but may not make payouts until far in the future (for example, life insurance, deferred annuities, or long term care insurance). This difference in the timing of liquidity and liabilities may distort the insurers portfolio choice (Thompson 2010, models endogenous counterparty risk) and raise the possibility of default or non-payment in immature markets. Assumption 1 allows for this insurer moral hazard without specifying their full portfolio problem.

It may also be the case that demand frictions are larger, in the sense of being more negative or shifted to the left, in immature markets. Status quo biases, distrust of new products, or inertia costs all imply smaller biases against products in mature markets compared with immature markets. As does lack of awareness (Boyer et al. 2020) or when new products are less likely to enter consideration sets. Similarly, negative demand frictions that disappear as consumers learn or experience the product¹⁴ can be represented in the same way. In this setting, immature markets have *more negative* demand frictions in contrast to mature markets where a larger fraction of consumers

¹⁴Evidence on tax minimization (Chetty et al. 2013) and retirement plan participation (Duflo and Saez 2003) suggests peer experience matters.

have been exposed to a product in the past through peers, advertising, or past purchase.

Instead of demand frictions shifting systematically between immature and mature markets, it may be the case that consumers beliefs about new products are noisier. This would be reflected in a greater dispersion of demand frictions in immature markets, which captures unbiased, but more diffuse, priors over the value of the insurance contract. Although it need not be the case that priors over new products are unbiased, this increase in noise is not ruled out by Assumption 1. Here, as long as the distribution of objective risk premia also moves between mature and immature markets, Assumption 1 is weaker than requiring stochastic dominance of objective risk premia, r , or demand frictions, ϵ . It only needs to be satisfied *on net*.

In contrast to [Spinnewijn \(2017\)](#), Assumption 1 does not require consumers are, on average, unbiased. Systematic optimism or pessimism in mature markets is not ruled out by Assumption 1. Even in established selection markets, such as annuities, there is substantial evidence of choice frictions ([Brown et al. 2021](#)) and biased beliefs ([O'Dea and Sturrock 2023](#)). Likewise positive systematic biases leading to over-purchase are compatible with Assumption 1, in this case it instead requires advertising or word-of-mouth (see [Bronnenberg et al. 2019](#), for a review) on average increases willingness-to-pay in mature markets.

Finally, to make the markets comparable, I hold the distribution of potential and realized risks constant across mature and immature markets. Assumption (2) formalizes that these are mature and immature markets for an identical insurance product with identical potential and realized risk pools.¹⁵

ASSUMPTION 2. Risk types, π_i , have identical marginal distributions in mature and immature markets.

These assumptions restrict how marginal distributions differ between mature and

¹⁵Neither firms nor individuals can alter the distribution of risks in mature markets.

immature markets. Yet, they do not capture how the sorting of consumers into insurance may change as markets mature. These differences in sorting can occur because of changes in consumers' self insurance behaviour or behavioural frictions. This sorting can improve the match of individuals to insurance plans, but also change how purchasing insurance signals an individual's cost. The following restriction on rank dependence capture differences in sorting patterns and generalize a restriction on correlation.

ASSUMPTION 3. *The dependence between π_i , and $\tilde{r}_i$ in mature and immature markets is represented by the bivariate copulas C^{IM} and C^{MM} . Their first derivatives satisfy $\frac{\partial C^{MM}(u,v)}{\partial x} \leq \frac{\partial C^{IM}(u,v)}{\partial x} \quad \forall u, v \in [0, 1], x \in \{u, v\}$.*

This orders the joint distribution *between* markets by their implied dependence. A higher rank along one dimension of a consumer's type is associated with a larger change in their joint probability in immature markets. Thus, the two types are more tightly distributed. Intuitively, this means that there is positive re-sorting between mature and immature markets and more dependence between the components of willingness-to-pay in immature markets. In economic terms, it rules out self insurance patterns that reverse risk exposure between mature and immature markets. This rules out settings in which early adopters are more risk averse and, consequently, more advantageously selected (see [Cutler et al. 2008](#) and [Fang et al. 2008](#) for empirical evidence on advantageous selection). Although this would be inconsistent with the motivation discussed thus far, the comparative static results below apply more generally. They hold for any pair of adversely selected markets that differ in their willingness-to-pay and sorting.¹⁶

Identical copulas, which imply no re-sorting effects, trivially satisfy Assumption 3. For common parametric copula families, Assumption 3 is a restriction on the parameter

¹⁶Along these line, similar results can be derived for an ordering of advantageously selected markets under an alternative version of Proposition 1.

space, for example a correlation ordering, $\text{corr}^{IM}(\pi, \tilde{r}) \geq \text{corr}^{MM}(\pi, \tilde{r})$, for normal copulas. More formally: larger quantiles of risk premia net of frictions, $\tilde{r}$, are associated with a larger increase in the cumulative density of costs in immature markets and vice-versa.

Lemma 2 formalizes the notion of re-sorting implied by Assumption 3.

LEMMA 2 (Increased Concordance). *The dependence between π_i and $\tilde{r}_i$ in mature and immature markets is represented by the bivariate copulas C^{IM} and C^{MM} . The dependence between cost type, π_i , and willingness-to-pay, ω_i , is represented by copulas $\hat{C}^{IM}$ and $\hat{C}^{MM}$.*

These copulas satisfy the following concordance ordering (Yanagimoto and Okamoto 1969; Tchen 1980):

$$C^{IM}(u, v) \geq C^{MM}(u, v) \quad \forall u, v \in [0, 1] \quad \text{and} \quad \hat{C}^{IM}(u, v) \geq \hat{C}^{MM}(u, v) \quad \forall u, v \in [0, 1]$$

This shows increased sorting on cost for two outcomes (see Appendix B for the proof). First, π and $\tilde{r}$ are larger in concordance in immature markets.¹⁷ Second, costs and willingness-to-pay are larger in concordance. Thus, the sorting on cost has increased. This second aspect of the result will be key to derive the implications below. This is because increased sorting on cost changes the extent to which the decision to purchase insurance provides a signal to the insurer. This then changes insurers' inference about the latent cost of the consumer at any given price.

While the results of this paper are derived under these general assumptions, closed form solutions are available under the assumption that copulas and marginals are normally distributed – which satisfy the conditions in Proposition 3 and Assumption 3. This increases transparency, but imposes strong additional restrictions, including allowing consumers to impose negative costs on the insurer and limiting tail risk.

¹⁷Both Anderson and Smith (2024) and Boerma et al. (2023) study concordance orderings as the outcome of assignment problems. Anderson and Smith (2024) show how the concordance order (which is identical to their positive quadrant dependence order with constant marginals) increases as the local synergies of the output function increase. Boerma et al. (2023) show that optimal assignments are larger in concordance when mismatch costs become less concave.

ASSUMPTION 4 (Joint Normality). π_i and $\tilde{r}_i$ are jointly normally distributed. Additionally, the variance of willingness-to-pay is constant, $\sigma_\omega^{IM} = \sigma_\omega^{MM}$, and correlation between risk type and willingness-to-pay is weaker in mature markets, $\text{corr}(\pi, \omega^{IM}) \geq \text{corr}(\pi, \omega^{MM}) > 0$.

To simplify exposition Assumption 4 rules out changes in the dispersion of willingness-to-pay across mature and immature markets.¹⁸ To increase accessibility in the main body of the text and to build intuition for the general results, I provide proofs of the paper's results under Assumption 4 for the special case of linear utility, $u(x) = x$, and provide proofs for the general non-parametric case in Appendix B.

2.2. Equilibrium in Mature and Immature Markets

I now turn to understanding the implication of Assumptions (1) - (3) on quantities, costs, and the market equilibrium.

First, at any given price, the average individual has a larger willingness-to-pay in mature markets and consequently more individuals choose to purchase insurance. Panel A of Figure 2 displays the implications of this *level effect* on demand and Proposition 2 formalizes it.

PROPOSITION 2 (Demand Contraction). *Willingness-to-pay is on average larger and demand expands in mature markets. $D^{IM}(p) \leq D^{MM}(p) \quad \forall p$*

PROOF. As the sum of the normal variables is itself normal, the demand curve is

$$D(p) = 1 - F_\omega(p) = 1 - \Phi\left(\frac{p - \mu_\omega}{\sigma_\omega}\right). \quad (9)$$

Assumptions (1) - (2) and (4) imply $\mu_\omega^{IM} \leq \mu_\omega^{MM}$ and σ_ω is constant, then distributions

¹⁸Under this condition, changes to the primitives take the form of location shifts for the marginal distributions. This is weaker than the stochastic ordering of normals as it does not impose a common variance across mature and immature markets for π_i and $\tilde{r}_i$.

Constant dispersion in willingness-to-pay is required for the normal distribution as the support is unbounded. Without this, equivalent results are available with truncated tails (Levy 1982).

of ω for mature markets first order stochastic dominate those of immature markets, $F_{\omega}^{MM}(p) \leq F_{\omega}^{IM}(p)$, which delivers the result. $\square$

This level effect on demand holds average and marginal costs constant for any given quantity. Hence, the cost curves faced by insurers in Panel A do not change. Although households need lower prices to induce purchase, the ordering of consumers willingness-to-pay is unchanged. Therefore the identity of marginal (and inframarginal) consumers at any quantity demanded is the same and, as Panel A shows the equilibrium price increases and demand falls (See Proposition 6).

Moreover, when there is *only* a level effect the utility function can be generalized to allow for arbitrary substitution patterns between π_i , and $\tilde{r}_i$. This is formalized in the following lemma with the proof given in Appendix B.

LEMMA 3 (Arbitrary Substitutes). *When there is no re-sorting ($C^{IM} = C^{MM}$), Proposition 2 can be extended to any willingness-to-pay satisfying*

$$\omega_i = u(\pi_i, \tilde{r}_i) \quad (10)$$

where $u(\cdot, \cdot)$ is a function increasing in its arguments.

Early adopters differ from the population of consumers who purchase in mature markets in important ways. First, they are more adversely selected as captured by the marginal cost of insurance.

PROPOSITION 3 (Increased Adverse Selection). *Consumers are adversely selected in both mature and immature markets. If resorting between markets satisfies the following additional assumption:*

- i. [Monotone Regression Dependence Ordering] $\tilde{g}_u^{IM}(g_u^{MM}(v))$ is increasing in v for all $u \in (0, 1)$, where $g_x(y) = \partial \hat{C}(x, y) / \partial y$ and $\tilde{g}_x \equiv g_x^{-1}$,

Then consumers are more adversely selected in immature markets
 $\left(\frac{\partial MC^{IM}(p)}{\partial p} \geq \frac{\partial MC^{MM}(p)}{\partial p} > 0 \right)$

PROOF. First, consider the joint (normal) distribution of cost type and willingness-to-pay,

$$\begin{pmatrix} \pi_i \\ \omega_i \end{pmatrix} \sim N \left[\begin{pmatrix} \mu_\pi \\ \mu_\omega \end{pmatrix}, \begin{pmatrix} \sigma_\pi^2 & \rho \sigma_\pi \sigma_\omega \\ \rho \sigma_\pi \sigma_\omega & \sigma_\omega^2 \end{pmatrix} \right],$$

where

$$\rho = \frac{\sigma_\pi^2 + \rho_{\pi,r} \sigma_\pi \sigma_r}{\sigma_\omega \sigma_\pi} > 0. \quad (11)$$

Then the marginal cost curve has the following closed form expression:

$$MC(p) = E(\pi | \omega = p) = \mu_\pi + \rho \frac{\sigma_\pi}{\sigma_\omega} (p - \mu_\omega), \quad (12)$$

which is increasing in p when ρ is positive. The second part of the proposition then follows from Assumption 4. $\square$

Intuitively, those who have the highest willingness-to-pay for insurance impose the highest costs on insurers for their coverage – making early adopters uniformly more costly at the margin. The proof in the general case follows from a direct application of Proposition 2.1 in Fang and Joe (1992).¹⁹

Unfortunately, it is difficult to express the additional assumption required in the general case as lower level restrictions on the joint distribution between cost type and risk premia (see Appendix A.1 for a derivation of the copula $\hat{C}$ using C). While this assumption lacks intuitive appeal, however, it is only required for this specific

¹⁹Again, for parametric families of copulas satisfying this assumption is typically a restriction on the parameter space. See Joe (2014) Chapter 4 for a systematic review.

result. This is because the usual definition of adverse selection invokes the slope of the marginal cost curve. While this is useful to distinguish adverse and advantageous selection, the notion of being *more* selected it gives rise to is too stringent. Instead, the characterisation of the equilibrium in this paper use results on how adverse selection determines average and marginal costs *in levels* and for a *given demand*. These are derived under the milder assumptions presented thus far.

Further comparing the marginal cost curves in mature and immature markets gives the following lemma relating changes in adverse selection to changes in sorting between markets.

LEMMA 4 (Adverse Selection In Levels). *When there is no re-sorting ($\hat{C} = \hat{C}^{IM} = \hat{C}^{MM}$) marginal costs are higher at any price. However, when re-sorting occurs, effects on marginal cost are ambiguous.*

The differences in marginal costs across markets depend on a level effect which holds consumer sorting fixed and captures differences in the level of demand at any price. However, while the slope of the marginal cost curve increases, the effect of re-sorting has an ambiguous effect on the level of marginal costs.

I make this formal using expression (12) for the marginal cost in the normal case:²⁰

COROLLARY 1 (Decomposing Marginal Costs). *Under Assumption 4, the differences in marginal costs between mature and immature markets can be decomposed into the level effect of demand and a re-sorting effect*

$$MC^{IM}(p) - MC^{MM}(p) = \frac{\sigma\pi}{\sigma_\omega} \rho^{IM} \left[\underbrace{(\mu_\omega^{MM} - \mu_\omega^{IM})}_{\text{level effect} \geq 0} + \underbrace{\left(1 - \frac{\rho^{MM}}{\rho^{IM}}\right) (p - \mu_\omega^{MM})}_{\text{re-sorting effect}} \right]. \quad (13)$$

²⁰It is always possible to define a cost curve holding sorting fixed, MC^{nosort} , and decompose the general case as $(MC^{IM}(p) - MC^{nosort}(p)) + (MC^{nosort}(p) - MC^{MM}(p))$. However, in the general case this does not reveal any intuitive insights.

This result makes the intuition for the preceding lemma explicit. Absent re-sorting, the increase in marginal costs depends only on the contraction in demand as the second term is equal to zero. However, when there is re-sorting the sign of the second term is ambiguous. For prices above the average willingness-to-pay in mature markets, re-sorting increases the difference in marginal costs between mature and immature markets. When prices are lower, however, the effect is negative as those purchasing insurance at high prices in immature markets are more adversely selected. Consequently, the pool of remaining consumers at low prices have lower costs.

Nevertheless, Proposition 4 highlights that, despite ambiguity at the margin, when consumers are reordered and selection on cost increases, then insurers face higher *average* costs.

PROPOSITION 4 (Cost Expansion). *Average costs are increasing in price and early adopters are higher cost on average, $AC^{IM}(p) \geq AC^{MM}(p)$.*

PROOF. Under joint normality, average costs are the expected value of a truncated normal distribution. The average cost curve is

$$AC(p) = E(\pi | \omega \geq p) = \mu_\pi + \rho \sigma_\pi \frac{\phi(\frac{p - \mu_\omega}{\sigma_\omega})}{1 - \Phi(\frac{p - \mu_\omega}{\sigma_\omega})}. \quad (14)$$

The average cost depends on the inverse Mills ratio of the subjective willingness-to-pay. This captures the extent to which selection truncates the distribution of cost types relative to its full support.

Comparing across markets gives $\forall p$

$$\frac{AC^{IM}(p) - AC^{MM}(p)}{\sigma_\pi \rho^{IM}} = \underbrace{\left(\frac{\phi(\frac{p - \mu_\omega^{IM}}{\sigma_\omega})}{1 - \Phi(\frac{p - \mu_\omega^{IM}}{\sigma_\omega})} - \frac{\phi(\frac{p - \mu_\omega^{MM}}{\sigma_\omega})}{1 - \Phi(\frac{p - \mu_\omega^{MM}}{\sigma_\omega})} \right)}_{\text{level effect} \geq 0} + \underbrace{\left(1 - \frac{\rho^{MM}}{\rho^{IM}} \right) \frac{\phi(\frac{p - \mu_\omega^{IM}}{\sigma_\omega})}{1 - \Phi(\frac{p - \mu_\omega^{IM}}{\sigma_\omega})}}_{\text{re-sorting effect} \geq 0} \geq 0 \quad (15)$$

The inverse Mills ratio and the difference in the inverse Mills ratio are always positive by the log concavity of the normal distribution ensuring non-negativity. $\square$

Again this expression can be decomposed into the contribution from the level and re-sorting effects. In contrast to marginal costs, both effects work in the same direction (raising average costs at a given price), because the effect of additional selection on costs among the inframarginal consumers dominates the ambiguous effect on the marginal consumer.

The first term in (15) captures changes in the level of demand, holding the consumer sorting constant, on the costs of early adopters. The second term in (15), which captures re-sorting, holds fixed the level of selection in immature markets and accounts for the additional sorting on costs which occurs.

The size of both effects are increasing in the share of heterogeneity in willingness-to-pay coming from cost type, π , while resorting effects depend on the amount of selection in immature markets. When this selection is large, insurance purchase is a stronger signal of cost. Consequently, increased sorting magnifies existing selection.

Panel (A) of Figure 2 shows how the decline in willingness-to-pay affects equilibrium price and quantity in the absence of an effect on costs. Crucially, although average costs increase at any price when there is a level effect, a change in the sorting patterns of individuals is necessary to induce a change in the cost curves insurers face conditional on quantity.

Suppose individuals were ordered by their willingness-to-pay. Without re-sorting, this ordering is identical in mature and immature markets. Thus when q individuals purchase insurance, the marginal consumer is identical (q along this order) and imposes identical expected costs on the insurer. If, instead, there is re-sorting, individuals are reshuffled and the identity of the marginal consumer changes. Increased sorting on cost in immature markets, relative to mature markets, results in this new marginal

consumer having higher expected costs. Purchase of the insurance signals higher costs in the immature market. Proposition 5 formalizes this intuition.

PROPOSITION 5 (Cost Expansion Fixing Demand). *Average and Marginal Costs as a function of quantity are larger in immature markets, $AC^{IM}(q) \geq AC^{MM}(q)$ and $MC^{IM}(q) \geq MC^{MM}(q)$. Absent re-sorting, they are identical.*

PROOF. Substituting inverse demand into (12) gives marginal costs as a function of quantity,

$$MC(q) \equiv E\left(\pi|\omega = D^{-1}(q)\right) = \mu_\pi + \rho\sigma_\pi\Phi^{-1}(1-q). \quad (16)$$

It follows that

$$MC^{IM}(q) - MC^{MM}(q) = \underbrace{(\rho^{IM} - \rho^{MM})(\sigma_\pi\Phi^{-1}(1-q))}_{\text{re-sorting effect}} \geq 0 \quad \forall q,$$

which shows that the marginal costs are higher at any quantity demanded.

Substituting inverse demand into (14) gives average costs as a function of the quantity demanded,

$$AC(q) \equiv E\left(\pi|\omega \geq D^{-1}(q)\right) = \mu_\pi + \rho\sigma_\pi \frac{\phi(\Phi^{-1}(1-q))}{1 - \Phi(\Phi^{-1}(1-q))} = \mu_\pi + \rho\sigma_\pi \frac{\phi(\Phi^{-1}(1-q))}{q}. \quad (17)$$

Differences in average costs are then

$$AC^{IM}(q) - AC^{MM}(q) = \underbrace{\rho^{IM}\sigma_\pi \left(1 - \frac{\rho^{MM}}{\rho^{IM}}\right) \frac{\phi(\Phi^{-1}(1-q))}{q}}_{\text{re-sorting effect}} \geq 0 \quad \forall q. \quad (18)$$

□

Panel B of Figure 2 shows this outward expansion of the average cost curve in

immature markets.²¹ Even holding the level of demand constant, the increase in adverse selection among the inframarginal consumers of the insurance product drives up expected costs for the insurer. Consequently, insurers set higher prices to cover the cost of providing insurance, even though some lower cost individuals select out of the pool. Thus, the equilibrium price increases and quantity falls. There is (additional) partial unraveling in the market.²²

PROPOSITION 6 (Partial Unraveling). *If an equilibrium exists, the equilibrium price is higher in immature markets and fewer consumers are insured, $q^{MM} \geq q^{IM}$ and $p^{MM} \leq p^{IM}$.*

PROOF. Let p^{MM} be the equilibrium price in the mature market,

$$D^{MM}(p^{MM}) = AC^{MM}(p^{MM}) . \quad (19)$$

If it is also an equilibrium price in the immature market, then

$$D^{IM}(p^{MM}) = AC^{IM}(p^{MM}) \quad (20)$$

also holds. But, from Propositions 2 and 4, the following inequality must also hold

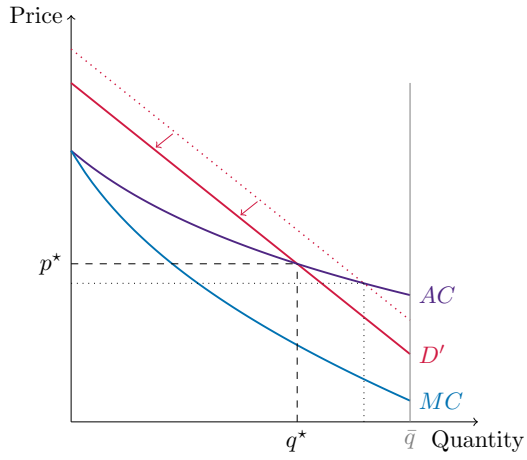
$$D^{IM}(p^{MM}) \leq D^{MM}(p^{MM}) = AC^{MM}(p^{MM}) \leq AC^{IM}(p^{MM}) , \quad (21)$$

which is a contradiction unless the immature and mature markets are identical.

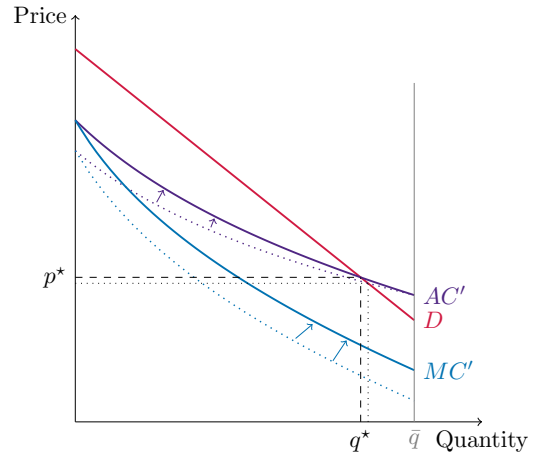
Furthermore, if there exists a price $p^{IM} \neq p^{MM}$ that clears the immature market

²¹This is a rotation of the average cost curve around full insurance as in Mahoney and Weyl (2017). The difference in average costs in (18) is decreasing in quantity demanded, reaching zero at full coverage. Intuitively, at full coverage the entire population is insured and $AC^{IM}(1) = AC^{MM}(1) = \mu\pi$. In contrast, Starc (2014) considers rotation around equilibrium demand.

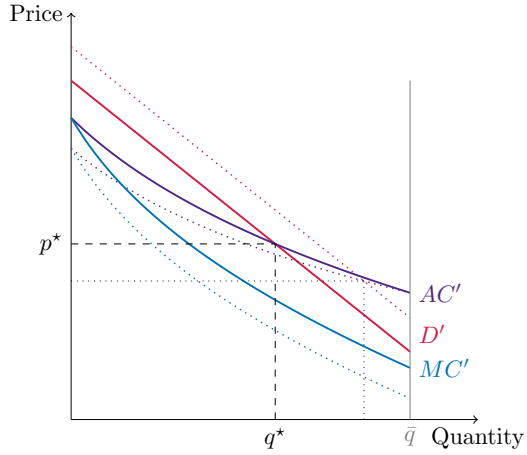
²²This can occur even if marginal costs in immature markets fall below demand which, absent choice frictions, implies full coverage is efficient.



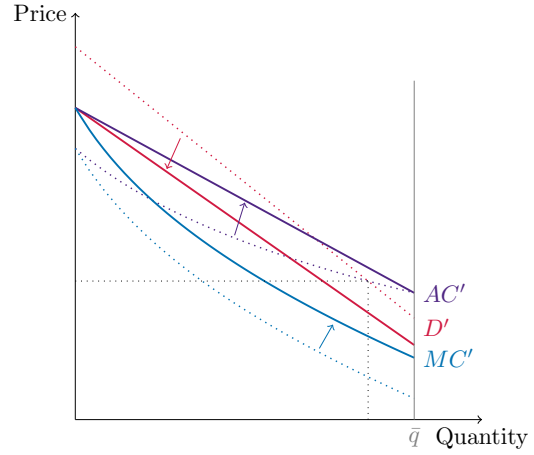
(A) Demand Contraction (Level Effect)



(B) Increased Adverse Selection (Sorting Effect)



(C) Under-insurance in the Short Run



(D) Unraveling in the Short Run

FIGURE 2. The Long and Short Run of Insurance Markets

then differences across markets satisfy *strict* dominance. Therefore

$$D^{IM}(p^{IM}) = AC^{IM}(p^{IM}) \longleftrightarrow p^{IM} > p^{MM}, \quad (22)$$

and

$$q^{IM} = D^{IM}(p^{IM}) < D^{MM}(p^{IM}) < D^{MM}(p^{MM}) = q^{MM}. \quad (23)$$

□

Panel C of Figure 2 illustrates the new equilibrium graphically. Marginal costs are below demand in both mature and immature markets, thus full insurance is the efficient outcome in both regimes. Despite this, there is additional under-insurance relative to the mature market due to both the level effect of demand contraction and the increased adverse selection due to additional sorting. The increase in equilibrium price and reduction in quantity demanded is larger than either effect in isolation (Panels A and B).

An alternative consequence of immaturity is that markets may unravel.

COROLLARY 2 (Complete Unraveling). *Would-be efficient mature markets can unravel before they reach maturity.*

This happens when the average cost curve lies above demand for all prices (Panel D of Figure 2). Thus, even a market that would be efficient if it could mature can fully unravel before maturing when the systematic differences between mature and immature demand (and the adverse selection that follows) are large enough.²³ Again, this can occur irrespective of whether under-insurance in immature markets is driven by rational differences in consumer valuations or irrational systematic biases. This abstracts from administrative loads and fixed costs for insurers, however, by increasing the costs of providing insurance at any given demand, these features reinforce this mechanism.

The presence of under-insurance ($q^{IM} < q^{MM}$) does not depend on whether changes in willingness-to-pay are driven by shifting risk premia or choice frictions. Calculating welfare costs, however, require a stance on their relative importance and the source of the change. The following corollary evaluates welfare under two policy relevant benchmarks.

²³Of course there may be markets that *should* never mature. This paper considers successful mature markets, yet the comparative static tools can identify ultimately unsuccessful mature markets ex ante.

COROLLARY 3 (Efficiency Costs). *Consumer surplus in immature markets is lower at the immature market equilibrium than the mature market equilibrium ($CS^{IM}(q^{IM}) \leq CS^{IM}(q^{MM})$). This is true irrespective of whether demand for insurance is evaluated at:*

- i. *Willingness-to-pay in immature markets (revealed preference benchmark)*
- ii. *Willingness-to-pay in mature markets (paternalistic benchmark)*

Thus, the equilibrium in immature markets is pareto dominated by the mature market.

PROOF. For a demand level q^* , consumer surplus in immature markets is defined as

$$CS^{IM}(q^*) = \int_0^{q^*} \tilde{D}^{IM}(q) - MC^{IM}(q) dq \quad \text{where} \quad \tilde{D}(q) = D^{-1}(q), \quad (24)$$

under the revealed preference benchmark. It follows that the difference in consumer surplus due to increased adverse selection is

$$CS^{IM}(q^{MM}) - CS^{IM}(q^{IM}) = \int_0^{q^{MM}} \tilde{D}^{IM}(q) - MC^{IM}(q) dq - \int_0^{q^{IM}} \tilde{D}^{IM}(q) - MC^{IM}(q) dq, \quad (25)$$

and given $q^{MM} \geq q^{IM}$ (proposition 6) this can be expressed as

$$CS^{IM}(q^{MM}) - CS^{IM}(q^{IM}) = \int_{q^{IM}}^{q^{MM}} \tilde{D}^{IM}(q) - MC^{IM}(q) dq \geq 0, \quad (26)$$

which captures the expanded coverage of insurance in mature markets. Similarly, for

the paternalistic benchmark which uses the mature market inverse demand

$$CS^{IM}(q^{MM}) - CS^{IM}(q^{IM}) = \int_{q^{IM}}^{q^{MM}} \tilde{D}^{MM}(q) - MC^{IM}(q) = \quad (27)$$

$$\underbrace{\int_{q^{IM}}^{q^{MM}} \tilde{D}^{MM}(q) - \tilde{D}^{IM}(q)}_{\text{Paternalism } (\geq 0)} + \underbrace{\int_{q^{IM}}^{q^{MM}} \tilde{D}^{IM}(q) - MC^{IM}(q)}_{\text{Case (i)}} \geq 0, \quad (28)$$

where the positive paternalistic contribution follows from Proposition 2.

Finally, note that firms always earn zero profits in equilibrium. Consequently, consumer surplus is sufficient to pareto order equilibria. $\square$

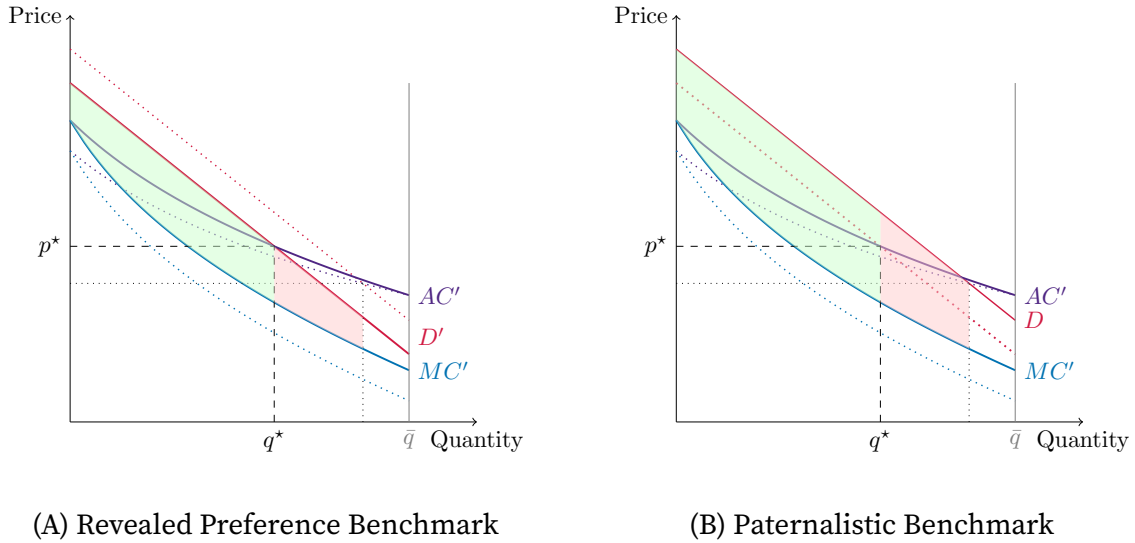


FIGURE 3. Welfare Losses in Immature Insurance Markets

Figure 3 illustrates the loss in consumer surplus due to increased adverse selection in immature markets. In both panels, the red shaded area to the right of q^* denotes the foregone surplus due to the additional under-insurance in the short run. Panel A considers the revealed preference benchmark in equation (26) which uses willingness-to-pay in immature markets as the welfare relevant payoff. Consequently,

foregone welfare is the area between the marginal cost curve and the demand curve in the immature market. Panel B instead shows the corresponding welfare loss under a paternalistic benchmark where the mature market willingness-to-pay is welfare relevant. Thus, the welfare loss is the area between the immature market marginal cost curve and demand in the mature market. The proof of Proposition 3 additionally shows that this welfare loss is larger under a paternalistic benchmark as willingness-to-pay expands in mature markets. As with equilibrium under-insurance, these results can be driven by a combination of the level effect of demand contraction and the increased adverse selection due to additional sorting. Yet, regardless of the mechanism, the efficiency cost of adverse selection is larger in immature markets due to additional under-insurance.²⁴

3. Evidence from the Market for Deferred Income Annuities

I now provide evidence that the dynamics of price, quantity, and selection in the market for deferred income annuities are consistent with the predictions of the stylized theory outlined above. I focus on the market for deferred income annuities – a product which operates like a traditional annuity income stream, but where payouts do not begin until a date (or age) agreed in the future. Hence, they provide deferred income.

Data on the aggregate size of the market, as measured by the dollar value of new annuity considerations, is collected from the quarterly reports of the Life Insurance Marketing and Research Association.²⁵ Figure 4 shows how the size of this market has evolved over time. At the start of this period the market for deferred income annuities was small, totaling around \$200 million and accounting for less than 1% of the overall

²⁴It is trivial to verify that this result combined with Proposition 5 implies that consumer surplus in mature markets is generically larger than immature markets.

²⁵Specifically, the U.S. Individual Annuity Sales Survey published by the LIMRA Secure Retirement Institute. Specific breakdowns for deferred income annuities are not available before 2012 as the value of the market is minimal.

fixed income annuity market. Subsequently, however, the market expanded rapidly between 2012 and 2014, with the real value of new considerations more than quadrupling in size. I use this market size as a proxy for market maturity.

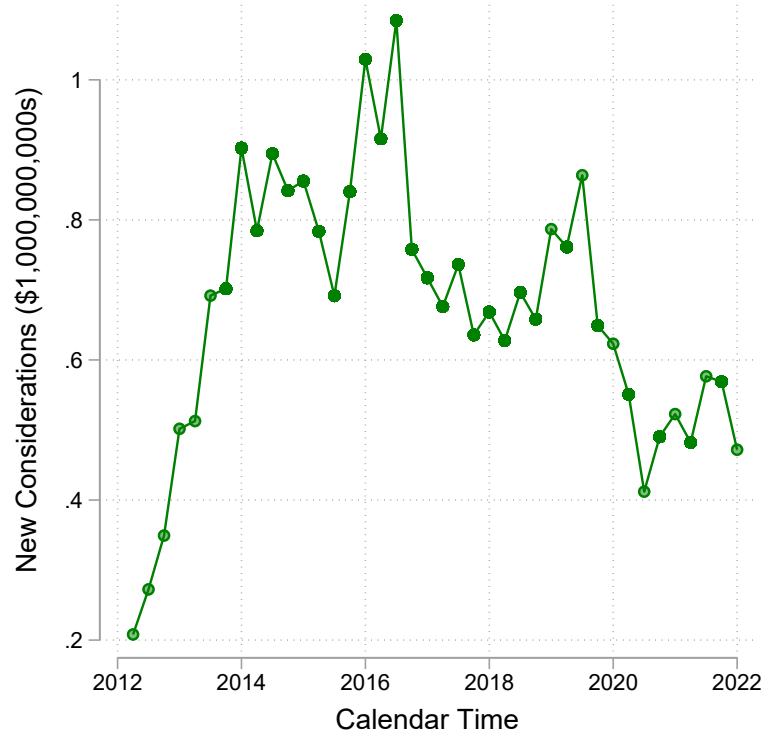


FIGURE 4. The Size of the Market for Deferred Income Annuities

To study the equilibrium relation between price, quantity, and selection I complement this data on aggregate quantities with quotes for individual annuity policies sourced from *Annuity Shopper* magazine between 2013 and 2021.²⁶ I focus on deferred income annuities bought at the age of 65. Each policy offered to annuitants of gender, g , is indexed by the identity of the insurance firm, i , and the deferral age, d , at which payouts begin. Over the sample, between 5 and 11 firms offer quotes for given policies at a point in time and quotes provide the annual payout offered per \$100,000 paid for the deferred income annuity. Figure 5 shows how the average payout for men

²⁶Before the third quarter of 2013 quotes for deferred income annuities are not available and, sadly for retirement interested researchers everywhere, annuity shopper ceased to operate in 2021.

(panel A) and women (panel B) changes over the sample period.

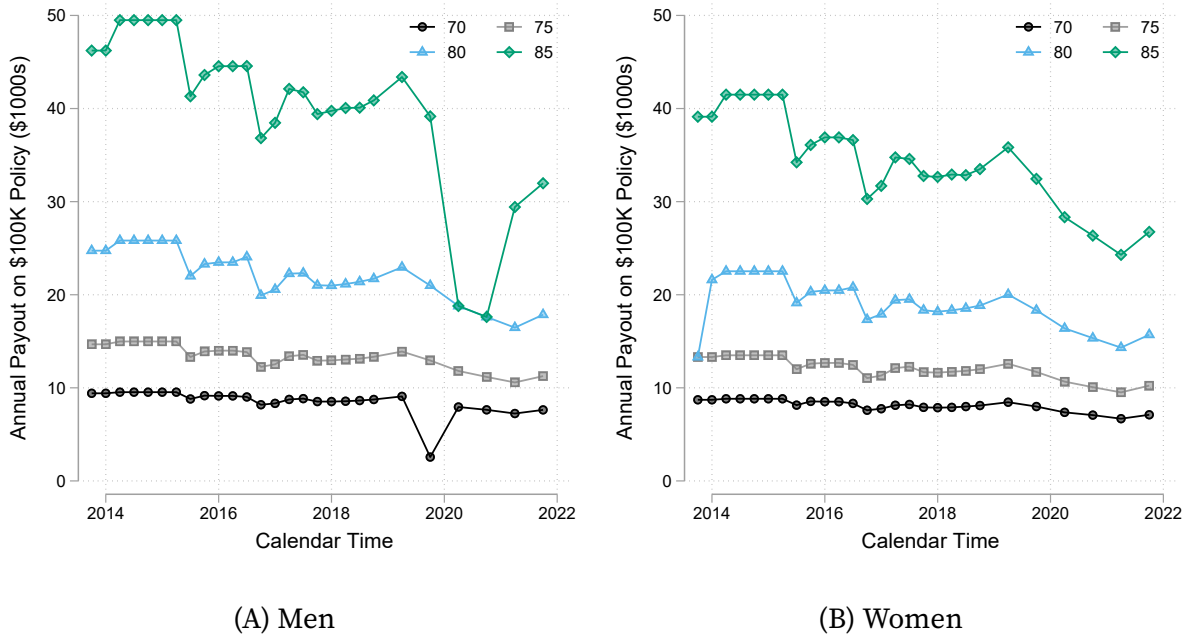


FIGURE 5. Average Payouts from Deferred Income Annuity Policies

Policies with later deferral dates, which expose consumers to greater mortality risk, offer larger payouts on average. Moreover, there are systematic differences across men and women consistent with differences in their expected longevity. In general, despite these systematic differences, average payouts share similar trends across policy characteristics. This is because the largest cost to the insurer is the opportunity cost of foregone investment (see [Mitchell et al. 1999](#)). To calculate the implicit price, and implied degree of adverse selection, I compute the expected present discounted value of the associated annuity income streams. The implied selection is then the per dollar deviation from the expected present discounted value under no adverse selection. For an annuity paying A per year after age d this is given by

$$Selection = 1 - \frac{PDV}{100K} = 1 - \frac{1}{100K} \sum_{a=65}^{110} \frac{A \times \mathbb{1}[a \geq d] \times S_a}{\prod_{j=a-64}^{110-64} (1 + i_j)}, \quad (29)$$

where a positive value indicates consumers experience less than actuarially fair pricing.

I use the BBB corporate bond yield at 10-year maturity to determine the insurer's interest rate, i_j , and cohort life tables for the age 65 population to determine mortality risk S_a at a given point in time. The deviation from actuarially fair pricing provides a measure of pricing and implied adverse selection for each contract. Given this measure of adverse selection for each contract, I estimate the partial correlation between implied selection at the policy level and aggregate quantity, $\overline{Q}_t$, while flexibly controlling for policy characteristics

$$Selection_{t,i,g,d} = \beta_0 + \delta_g + \delta_i + \delta_d + \beta_1 \overline{Q}_t + u_{t,i,g,d}. \quad (30)$$

Here it is important to stress that this is a comparison of how equilibrium pairs of selection (price) and quantities co-move. This is not an attempt to recover demand (or cost) curves from equilibrium pairs.²⁷ Instead, these pairs of equilibrium selection and aggregate quantity allow for a direct test of the empirical content of the theoretical model's equilibrium predictions. Are mature markets, proxied by market size, associated with less selection on average as the theory predicts?

Table 1 reports the estimated value of β_1 , the coefficient of interest, across three different specifications. The first column controls only for the gender of annuitants and reports the partial correlation. Absent policy characteristics, a \$1 billion increase in the size of the market implies an almost 5 cent reduction in the implied adverse selection on each dollar of annuities purchased. Columns two and three repeat this analysis with additional controls for deferral age and insurer identity. Finally, column four includes all controls. These additional fixed effects account for systematic differences in the consumers of plans by deferral age and insurer (for example due to geographic differences in insurer market penetration). While the overall explanatory power of

²⁷This would of course be subject to the usual concern of simultaneity.

	(1)	(2)	(3)	(4)
$\overline{Q}_t$ (Billions)	-0.048*** (0.017)	-0.047*** (0.011)	-0.046*** (0.018)	-0.044*** (0.012)
<i>FEs</i>				
Gender	Yes	Yes	Yes	Yes
Deferral Age	No	Yes	No	Yes
Insurer	No	No	Yes	Yes
N	1, 748	1, 748	1, 748	1, 748

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Huber-White heteroskedasticity robust standard errors are reported. The estimated partial correlation corresponds to the coefficient β_1 in equation (30). Implied selection is recovered using equation (29) and data collected from policy quotes printed in *Annuity Shopper* magazine. Aggregate quantities correspond to the real value of new deferred income annuity considerations collected from LIMRA and reported in Figure 4.

TABLE 1. The Relationship Between Equilibrium Selection and Market Size

the regression increases, the point estimate is similar across all specifications and statistically significant. This exercise highlights that in this market adverse selection declines in mature markets.

4. Will Deep Pocketed Insurers Enter When Immature Markets Unravel?

Section 2 establishes that differences between mature and immature markets can lead to the unraveling of competitive equilibria when market immaturity exacerbates adverse selection. This naturally raises the question of whether dynamic incentives counteract these forces. To answer this, I illustrate the key economic forces determining insurer entry into new markets using a two firm, two period model. I show that, when adverse selection is severe enough, even the promise of market power is not enough to entice insurers to enter. Despite the fact that mature markets may be efficient, *temporary and*

predictable increases in adverse selection give rise to missing markets.

These results reverse the standard intuition for the interaction between market power and adverse selection derived from static frameworks. In static analysis (e.g. [Starc 2014](#) or [Mahoney and Weyl 2017](#) when quantity is low), adverse selection ameliorates the problems of market power by limiting the extent to which firms can earn rents by abusing market power. Market power, however, exacerbates under-supply. In contrast, this dynamic setting illustrates how market power can ameliorate dynamic inefficiencies stemming from adverse selection. This highlights the complexity of policy intervention in selection markets.

In each period, firms choose whether to offer an insurance product taking as given contract terms. They observe entry decisions and post prices, competing a la Bertrand. Consequently, prices equal average costs if both firms enter (equation 5). If, however, only one firm enters, they are able to exploit their market power and equate marginal revenue with marginal cost, delivering the usual monopolist pricing rule.

To close the model, I assume the market begins in an immature state. This state of the world updates from immature to mature in the second period if and only if at least one firm enters in the first period. This tractable framework captures the interaction between market power, adverse selection, and pricing, as well as that firms who want the market to reach maturity are engaged in a war of attrition with consumers.²⁸

The following proposition characterizes how market immaturity interacts with market forces and can endogenously give rise to market incompleteness in this setting. The proof is in Appendix C.

PROPOSITION 7. *If a monopolist cannot earn positive profits in immature markets, there exists a subgame perfect equilibrium where no firm enters. Moreover, this is the unique extensive-form trembling hand perfect equilibria.*

²⁸Consumers learn about the generic “product” offered by firms instead of developing brand loyalty.

Importantly, the statement depends only on primitives of the immature market and, thus, holds even when the mature market is efficient. In other words, dynamic incentives do not preclude missing efficient mature markets. Therefore, this rationalizes the absence of better insurance products including redesigned long-term care insurance or reverse mortgage products.

The intuition for the result is simple, but reveals a new free-riding problem preventing insurance markets from maturing. When consumer valuations of insurance contracts in immature markets are different from mature markets, those who wish to purchase insurance are adversely selected. When adverse selection is sufficiently large, a firm entering into an immature market is effectively investing (by earning negative profits) in the market reaching maturity.²⁹ If they could guarantee monopolist profits in the mature market, it may be profitable to absorb the temporary hit to their bottom line. Herein lies the free-rider problem. The entrant pays a cost to familiarize consumers with the new product, but another firm always has an incentive to wait and enter the mature market. Consequently, no firm is willing to enter the immature market.³⁰

Crucially, with adverse selection there is no guarantee positive profits exist – even for a monopolist. To see this, consider the following standard expression for the insurer’s profit margin (Starc 2014; Kong et al. 2023):

$$p - AC(p) = \frac{1}{\eta_{D,p}} - (AC(p) - MC(p)), \quad (31)$$

where the first term is the usual Lerner markup and the second term captures the influence of adverse selection on firm pricing through their desire to “cream skim”

²⁹Indeed, at least one major Canadian auto-insurer offers individuals a subsidy (above rating incentives) to enrol in new contracts using telematic monitoring.

³⁰This implicitly assumes that the aggregate state transitions only once an amount of contracts exceeds a positive threshold, thus firms cannot ‘enter’ by charging a price that guarantees zero-profits and no contracts sold.

low cost consumers at the margin (Mahoney and Weyl 2017, highlight that this effect is reversed at high quantities). Thus, adverse selection can discipline market power. However, when either the price elasticity of demand is large or adverse selection is severe enough, firms may earn negative monopolist profits in the absence of fixed costs.³¹

Implications For Market Intervention: Patents or Burdensome Regulation as Time Varying Subsidies. If there is surplus in mature markets, it may be socially optimal for a market to mature even if there is no-trade in its infancy. However, if primitives of markets change over time, permanent policy solutions, like a subsidy, may be inefficient. For instance, in the entry and exit game above there is a welfare cost associated with missing mature markets, but once the market reaches maturity the competitive allocation is efficient.

In this case, phasing out subsidies over time when there is deadweight loss from government intervention in mature markets is optimal. However, implementing a time varying subsidy may impose unreasonable information requirements on a social planner. Tautologically, there is no choice data to observe if a market is missing.

A simple decentralized alternative policy is to remove barriers to patenting or to increase the length of patent protection in insurance markets. Intuitively, this raises the monopolist profit in immature markets by protecting firms against free riders and encourages entry. It offers an effective time-varying subsidy. In principal, this simplifies the policy problem because it now depends only on patent length (as opposed to a time varying subsidy schedule). However, there still remains an information problem: the planner needs to have enough information to select an optimal patent length.

One advantage of patents over subsidies is that the filing of patents requires

³¹Kong et al., similarly discuss entry under imperfect competition and derive a threshold condition on the slope of the average cost curve to sustain a given market structure.

information disclosure on the part of firms. This, in principal, can be used to collect information which can be used in determining the optimal patent length.³² In contrast, in the absence of the insurance market, a planner relying on subsidies has to will the insurance product into existence. This involves choosing from an abstract menu of hypothetical insurance ideas and hoping that it is technologically feasible. In contrast, the patenting planner is able to choose from a menu of observed innovations.

An alternative policy tool is to increase the regulatory burden on insurers (or comparable institutions in other selection markets) who offer new products. For example, regulatory agencies can require extensive disclosure requirements or mandate costly reporting of newly issued policies. This increases the cost of adopting existing technology and slows diffusion across firms (Benhabib et al. 2021).³³ This slowed diffusion lengthens the period of market power.

5. Discussion

This paper derives general comparative static results for the equilibrium of selection markets under direct assumptions on the joint density of model primitives. It allows for multi-dimensional heterogeneity and develops a partial ordering of markets. Markets are ordered by the primitives of their economic environment and I show how this delivers an ordering over equilibrium prices, quantity, selection, and welfare that can be represented using the popular graphical framework for insurance analysis (Einav and Finkelstein 2011).

I use these tools to study the competitive equilibrium in mature and immature markets for identical insurance contracts. Under economically motivated and plausible assumptions on risk premia and the potential biases of consumers, early adopters

³²See Williams (2017) for a recent survey of other relevant considerations in the optimal design of patent length and scope.

³³Empirical evidence highlights the role of labor regulation in suppressing innovation. For example, Griffith and Macartney (2014) and Aghion et al. (2023).

are more adversely selected and impose higher costs on insurers even though the distribution of potential costs to the insurer are constant. I show that this leads to increased under-insurance and lower welfare in immature markets.

I use these results to understand missing markets, where firms do not offer potentially valuable insurance contracts. This extends the classic study of market unraveling in insurance markets to consider this dynamic form of market failure. This jointly explains low utilization, slow demand for new products, and market inactivity. Furthermore, I embed these results within a simple model of firm exit and entry to understand the benefits of patenting in insurance markets. This highlights a novel free-rider problem and shows that, when market primitives are dynamically evolving, the usual interplay between adverse selection and market power is reversed.

While I apply these tools to the study of insurance markets, the results apply more generally for a class of selection markets – where consumers can impose heterogeneous costs on the seller when they choose to purchase – under multidimensional heterogeneity. Crucially, they do not depend on whether market immaturity affects the true welfare value of insurance or only consumer’s perception of the value. It thus encompasses a variety of applications in economics. These same tools, for example, can be used to understand why shifts in risk premia lower adverse selection and derive sharp predictions on the lemons penalty as in [Blundell et al. \(2019\)](#).

Finally, the results in this paper emphasize that in addition to intra-temporal concerns, there is an inter-temporal motive for intervention in selection markets. It is crucial, therefore, to consider the welfare of consumers over the full transition path from immature to mature markets as well as drawing on estimates from a single point in time.

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Online Appendix

Appendix A. Proofs & Additional Results for Section 1

PROOF OF PROPOSITION 1. Proof of Part (i). Beginning with the monotonicity of marginal costs – the CDF of the conditional random variable $\pi|\omega = p$ is

$$\begin{aligned} F_{\pi|\omega}(\pi|\omega) &\equiv \int_{-\infty}^{\pi} f_{\pi|\omega}(u|\omega) du = \int_{-\infty}^{\pi} \frac{f_{\pi,\omega}(u, \omega)}{f_{\omega}(\omega)} du \\ &= \widehat{C}_2(u, F_{\omega}(\omega)) = g(F_{\pi}(\pi)), \end{aligned} \quad (\text{A1})$$

for a given copula $\widehat{C}$ where $\widehat{C}_2(x, y) = \partial \widehat{C}(x, y) / \partial y$.

As the marginal cost curve is a conditional expectation $MC(p) = E(\pi|\omega = p)$, stochastic dominance of the conditional distributions is a sufficient condition to order marginal costs in price

$$F_{\pi|\omega}(\pi|\omega = v_1) \succeq F_{\pi|\omega}(\pi|\omega = v_2) \quad \forall v_1 \geq v_2 \quad (\text{A2})$$

$$\longrightarrow MC(v_1) = E(\pi|\omega = v_1) \geq E(\pi|\omega = v_2) = MC(v_2) \quad \forall v_1 \geq v_2. \quad (\text{A3})$$

Substituting equation (A1) gives the following necessary and sufficient condition:

$$\widehat{C}_2(u, F_{\omega}(v_1)) \leq \widehat{C}_2(u, F_{\omega}(v_2)) \quad \forall v_1 \geq v_2 \text{ and } \forall u \quad (\text{A4})$$

$$\longleftrightarrow \widehat{C}_2(u, a) \leq \widehat{C}_2(u, b) \quad \forall a \geq b \text{ and } \forall u \quad (\text{A5})$$

as $v_1 \geq v_2 \longrightarrow F_{\omega}(v_1) \geq F_{\omega}(v_2)$. This is satisfied for all copulas where $C_2(\cdot, \cdot)$ is decreasing in its second argument. Condition (i) of Proposition (1) guarantees $\widehat{C}$ is concave and satisfies this property.

This establishes that $MC(p)$ is increasing in p .

Proof of Part (ii). Turning to the monotonicity of average costs – the CDF of the conditional random variable $\pi|\omega > p$ is

$$F_{\pi|\omega > p}(\pi|\omega > p) \equiv \frac{u - \widehat{C}(u, F_\omega(p))}{1 - F_\omega(p)}, \quad (\text{A6})$$

for a given copula $\widehat{C}$.

As the average cost curve is defined as a conditional expectation $AC(p) = E(\pi|\omega \geq p)$, stochastic dominance of the conditional distributions is a sufficient condition to order average costs in price

$$F_{\pi|\omega \geq p}(\pi|\omega \geq v_1) \succeq F_{\pi|\omega \geq p}(\pi|\omega \geq v_2) \quad \forall v_1 \geq v_2 \quad (\text{A7})$$

$$\longrightarrow AC(v_1) = E(\pi|\omega \geq v_1) \geq E(\pi|\omega \geq v_2) = AC(v_2) \quad \forall v_1 \geq v_2 \quad (\text{A8})$$

Substituting equation (A6) gives the following necessary and sufficient condition:

$$F_{\pi|\omega \geq p}(\pi|\omega \geq v_1) \leq F_{\pi|\omega \geq p}(\pi|\omega \geq v_2) \quad \forall v_1 \geq v_2 \text{ and } \forall \pi \quad (\text{A9})$$

$$\frac{u - \widehat{C}(u, F_\omega(v_1))}{1 - F_\omega(v_1)} \leq \frac{u - \widehat{C}(u, F_\omega(v_2))}{1 - F_\omega(v_2)} \quad \forall v_1 \geq v_2 \text{ and } \forall u \quad (\text{A10})$$

as $v_1 \geq v_2 \longrightarrow F_\omega(v_1) \geq F_\omega(v_2)$, condition (ii) of Proposition (1) guarantees $\widehat{C}$ satisfies this property, delivering an ordering of average costs for a given copula.

□

PROOF OF LEMMA 1. It is widely known that monotone likelihood ratio dependence is a stronger dependence property than those implied by the conditions in Proposition 1 (see, e.g., Nelsen 2006 Theorem 5.2.19).

□

A.1. Additional Results and Discussion of Copula

This section shows how to derive the distribution of willingness-to-pay directly from assumptions on cost and risk premia. This will depend on the dependence between cost and risk premia, as encoded by the copula C , and marginal distributions. The following results are derived in [Cherubini et al. \(2011\)](#) who also provide extensions to multivariate settings (see also [Navarro and Sarabia 2022](#)). These multivariate extensions can also be used to derive expressions for the joint-density of net premia from the underlying densities of risk premia r and choice frictions ε . This extends general results on the convolution of probability distributions.

The distribution function for the ω is given by F_ω

$$F_\omega(\omega) = \int_{-\infty}^{\infty} f_\pi(c) \partial_1 C(F_\pi(c), F_{\tilde{r}}(\omega - c)) dc \quad (\text{A11})$$

provided that $\lim_{v \rightarrow 0^+} \partial_1 C(u, v) = 0$ for all $u \in (0, 1)$. This is a mixture over the conditional distribution of $(\tilde{r} + c | \pi = c)$, which is given by³⁴

$$F_{\omega|\pi}(\omega | \pi = c) = \partial_1 C(F_\pi(c), F_{\tilde{r}}(\omega - c)). \quad (\text{A12})$$

We can use this to derive the following joint density function

$$F_{\pi, \omega}(t, v) = \int_{-\infty}^t F_{\omega|\pi}(v|t) dF_\pi(t) = \int_{-\infty}^t \partial_1 C(F_\pi(t), F_{\tilde{r}}(v - t)) dF_\pi(t), \quad (\text{A13})$$

which in turn can be used recover $\hat{C}$, the copula specifying the dependence between ω and π , by application of Sklar's Theorem:

$$\hat{C}(u, v) = \int_{-\infty}^v \partial_1 C(x, F_{\tilde{r}}(F_\omega^{-1}(u) - F_\pi^{-1}(x))) dx. \quad (\text{A14})$$

³⁴Theorem 2.3 [Navarro and Sarabia 2022](#) shows that this admits a distortion function representation.

In this general form, there are not closed form results for this integral as it depends on the marginal densities and the copula. However, we can additionally derive the following probability density functions which will prove useful in evaluating its properties:

$$f_{\omega|\pi}(\omega|\pi = c) = f_{\tilde{r}}(\omega - c)\partial_{1,2}C(F_{\pi}(c), F_{\tilde{r}}(\omega - c)) \quad (\text{A15})$$

$$f_{\omega}(\omega) = \int_{-\infty}^{\infty} f_{\pi}(c)f_{\tilde{r}}(\omega - c)\partial_{1,2}C(F_{\pi}(c), F_{\tilde{r}}(\omega - c)) dc \quad (\text{A16})$$

and the following corollary uses these representations to re-express Lemma 1 in terms of the joint distributions of π and $\tilde{r}$.

COROLLARY A1 (MLRP). *The monotone-likelihood ratio property for willingness-to-pay, ω , and cost type, π , is equivalent to the following property for the cost types and risk premia $\tilde{r}$:*

$$\frac{f_{\tilde{r}}(r-\pi)\partial_{1,2}C(F_{\pi}(\pi), F_{\tilde{r}}(r-\pi))}{f_{\tilde{r}}(r-\pi')\partial_{1,2}C(F_{\pi}(\pi'), F_{\tilde{r}}(r-\pi'))} \geq \frac{f_{\tilde{r}}(r'-\pi)\partial_{1,2}C(F_{\pi}(\pi), F_{\tilde{r}}(r'-\pi))}{f_{\tilde{r}}(r'-\pi')\partial_{1,2}C(F_{\pi}(\pi'), F_{\tilde{r}}(r'-\pi'))} \quad \forall \pi' \geq \pi \text{ and } r' \geq r$$

Thus, all joint distributions satisfying this property satisfy conditions (i) and (ii) in Proposition 3.

I now turn to discussing examples of common parametric copulas and the conditional distributions of ranks (summarized by their derivatives). While this allows for convenient theoretical analysis, it is worth emphasizing, however, that in empirical settings copulas can be estimated directly from the data using non-parametric tools. This is an inexhaustive list and other copula family also abound.^{35,36}

Example 1: Gaussian copula. The Gaussian copula is a family of copulas that vary in their correlation parameter ρ . All multivariate Normal distributions have Gaussian

³⁵Moreover, the convolution of copulas above is closed under mixtures which, with sufficient mixing, allows for almost arbitrary dependence using parametric families.

³⁶A less common copula in statistical applications, but highly relevant in economic applications is the Burr copula. This is characterized by joint-Burr distributions which exhibit similar jointness properties to the normal distribution. Moreover, univariate Burr distributions have extensive use as a generalized version of the Pareto distribution to model income distributions with long tails (see, for instance, Singh and Maddala 1976).

copulas and Normal marginals. However, not all normally distributed marginals have Gaussian copulas. The bivariate copula is defined as:

$$C(u, v; \rho) = \Phi^{-1}(u) \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\Phi^{-1}(v)} \exp \left[-\frac{1}{2} \frac{y^2 - 2\rho\Phi^{-1}(u)y + \Phi^{-1}(u)^2}{1-\rho^2} \right] dy \quad (\text{A17})$$

$$= \Phi^{-1}(u) \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{\Phi^{-1}(u)^2}{2} \right] \Phi \left[\frac{\Phi^{-1}(v) - \rho\Phi^{-1}(u)}{\sqrt{1-\rho^2}} \right] \quad (\text{A18})$$

with first derivative given by (it is symmetric)

$$\frac{d}{du} C(u, v; \rho) = \frac{d}{du} \Phi^{-1}(u) \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{\Phi^{-1}(u)^2}{2} \right] \Phi \left[\frac{\Phi^{-1}(v) - \rho\Phi^{-1}(u)}{\sqrt{1-\rho^2}} \right] \quad (\text{A19})$$

$$= \Phi \left[\frac{\Phi^{-1}(v) - \rho\Phi^{-1}(u)}{\sqrt{1-\rho^2}} \right], \quad (\text{A20})$$

and cross-partial derivative (or copula density)

$$\frac{\partial^2}{\partial u \partial v} C(u, v; \rho) = \frac{1}{\sqrt{1-\rho^2}} \frac{d}{dv} \Phi^{-1}(v) \Phi' \left[\frac{\Phi^{-1}(v) - \rho\Phi^{-1}(u)}{\sqrt{1-\rho^2}} \right] \quad (\text{A21})$$

$$= \frac{1}{\sqrt{1-\rho^2}} \exp \left[\frac{-\rho^2\Phi^{-1}(v)^2 + 2\rho\Phi^{-1}(u)\Phi^{-1}(v) - \rho^2\Phi^{-1}(u)^2}{2(1-\rho^2)} \right]. \quad (\text{A22})$$

Substituting into Equation A12 gives the following expression for the joint density in this example:

$$F_{\omega|\pi}(\omega|\pi = c) = \Phi \left[\frac{\Phi^{-1}(F_{\pi}(c)) - \rho\Phi^{-1}(F_{\tilde{r}}(\omega - c))}{\sqrt{1-\rho^2}} \right], \quad (\text{A23})$$

which highlights that the normality of the conditional density is a special property. It requires both the specific Gaussian copula and the normally distributed marginals.

Figure A1 shows how the parametric specification of the copula effects the joint distribution when the marginal distribution are fixed. The left columns shows how the joint distribution of π and $\tilde{r}$ changes with the correlation parameter. In both cases I

assume π and $\tilde{r}$ are log normally distributed with their logs following standard normals. Comparing panels A and C shows the role of the correlation parameter for the gaussian copula. As the correlation increases, it stretches the density towards the upper right corner.

Example 2: Gumbel Copula. The Gumbel copula has heavier tails than the normal distribution and is also asymmetric, with more weight in the right tail. It's left tail, however, is similar to the normal distribution. The bivariate copula is defined (for $\delta \geq 1$) as:

$$C(u, v; \delta) = \exp \left(- \left[(-\log u)^\delta + (-\log v)^\delta \right]^{1/\delta} \right) \quad (\text{A24})$$

$$= \exp \left(-A^{1/\delta} \right) \quad \text{where} \quad A = (-\log u)^\delta + (-\log v)^\delta \quad (\text{A25})$$

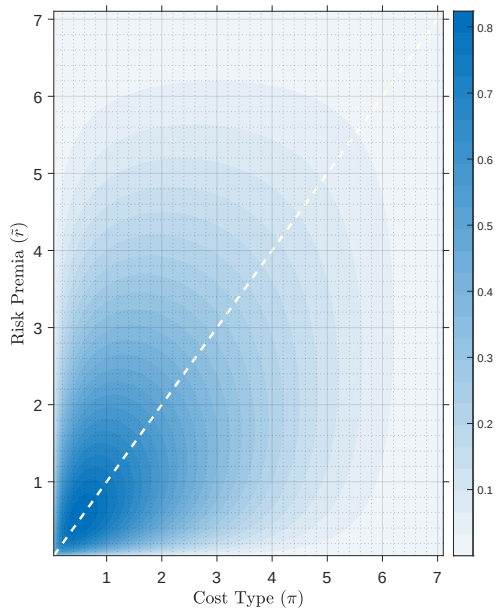
with first derivative given by (it is symmetric)

$$\frac{\partial}{\partial u} C(u, v; \delta) = \exp \left(-A^{1/\delta} \right) \cdot \frac{(1 + (-\log v / -\log u)^\delta)^{1/\delta - 1}}{u} \quad (\text{A26})$$

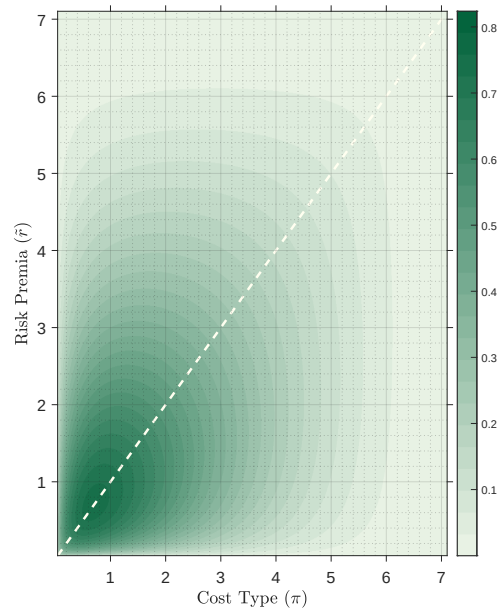
and cross-partial derivative

$$\frac{\partial^2}{\partial u \partial v} C(u, v; \delta) = \frac{(-\log v)^{\delta-1}}{v} \left[\frac{1-\delta}{\delta} (A^{1/\delta} + \delta - 1) A^{1/\delta - 2} \right]. \quad (\text{A27})$$

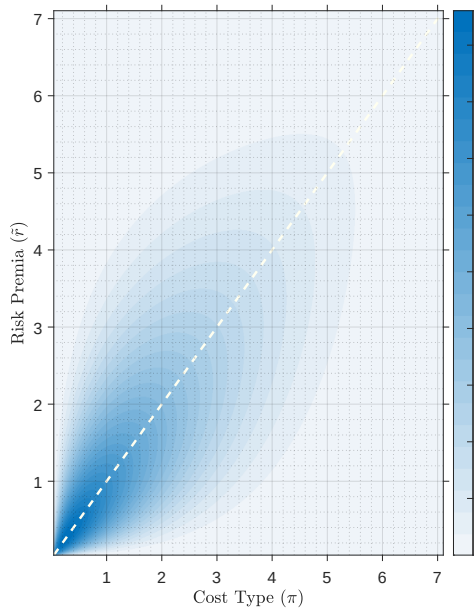
Comparing the right column of Figure A1, which shows the Gumbel copula, to the left highlights differences induced in the joint density by choice of the copula. At lower correlations, panels A and B, the two copulas deliver more similar joint-densities. However, panel B, using the Gumbel copula, reveals heavier tails of the joint distribution. This is even more pronounced for higher correlations shown in panels C and D. Here, like the Gaussian copula, panel D stretches the density towards the upper-right corner



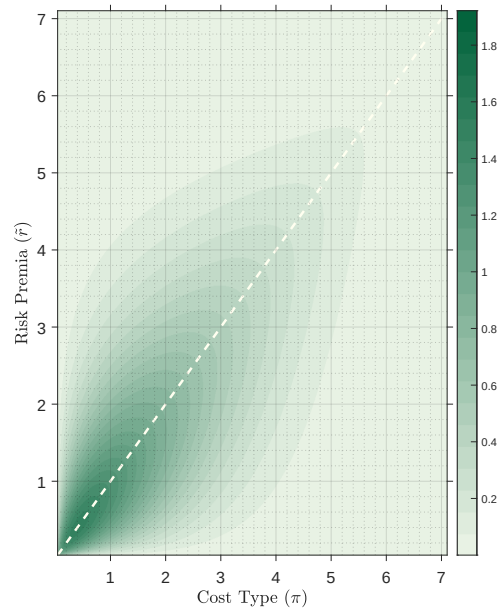
(A) Gaussian Copula (correlation=0.5)



(B) Gumbel Copula (correlation=0.5)



(C) Gaussian Copula (correlation=0.85)



(D) Gumbel Copula (correlation=0.85)

FIGURE A1. Copula and Joint Densities

Notes: Panels vary the copula (and parameter controlling correlation) for the rank dependency between cost type (π) and risk premia ($\tilde{r}$). In all panels the marginal distributions of cost type (π) and risk premia ($\tilde{r}$) are distributed log-normally with a mean of 0 and unit variance.

of the figure. In contrast though, the normal exhibits softer contours. Instead, the surface of the joint density is pulled “tauter” under the Gumbel copula and more mass is concentrated in the joint tail. Thus the risk is more jointly heavily tailed.

Example 3: The Independent Copula. The independent copula describes the case where the two random variables are independent. The relevant formulae are:

$$C(u, v) = uv \tag{A28}$$

$$\frac{\partial}{\partial u} C(u, v) = v \tag{A29}$$

$$\frac{\partial^2}{\partial u \partial v} C(u, v) = 1. \tag{A30}$$

Appendix B. General Proofs for Section 2

PROOF OF LEMMA 2. Integrating the inequality on the first derivatives in Assumption 3 gives:

$$\int \frac{\partial C^{MM}(u, v)}{\partial x} dx \leq \int \frac{\partial C^{IM}(u, v)}{\partial x} dx \quad \forall u, v \in [0, 1] \quad (\text{A31})$$

$$\rightarrow C^{MM}(u, v) \leq C^{IM}(u, v) \quad \forall u, v \in [0, 1], \quad (\text{A32})$$

which establishes the concordance ordering for π and $\tilde{r}$.

Using the equivalence of the copula concordance order and positive quadrant dependence ordering of the distribution function, establishing a concordance ordering for $\hat{C}$ across markets is equivalent to the following property of the joint distributions, $H_{\pi, \omega}(\cdot, \cdot)$, for all π and ω :

$$\int_{-\infty}^t F_{\omega|\pi}^{MM}(\omega|t) dF_{\pi}(t) = H_{\pi, \omega}^{MM}(t, \omega) \leq H_{\pi, \omega}^{IM}(t, \omega) = \int_{-\infty}^t F_{\omega|\pi}^{IM}(\omega|t) dF_{\pi}(t). \quad (\text{A33})$$

Inequalities are preserved under convex combinations, thus the following condition holding for all $t \in \text{supp } \pi$ and ω is sufficient for the inequality in Equation A33:

$$F_{\omega|\pi}^{MM}(\omega|t) = \partial_1 C^{MM}(F_{\pi}(t), F_{\tilde{r}}^{MM}(\omega - t)) \leq \partial_1 C^{IM}(F_{\pi}(t), F_{\tilde{r}}^{IM}(\omega - t)) = F_{\omega|\pi}^{IM}(\omega|t), \quad (\text{A34})$$

with

$$\partial_1 C^{MM}(F_{\pi}(t), F_{\tilde{r}}^{MM}(\omega - t)) \leq \partial_1 C^{IM}(F_{\pi}(t), F_{\tilde{r}}^{MM}(\omega - t)) \leq \partial_1 C^{IM}(F_{\pi}(t), F_{\tilde{r}}^{IM}(\omega - t)). \quad (\text{A35})$$

The first inequality holds for all $t \in \text{supp } \pi$ and ω under Assumption 3. Assumption 1 guarantees $F_{\tilde{r}}^{MM}(\tilde{r}) \leq F_{\tilde{r}}^{IM}(\tilde{r})$ and, thus, the second inequality is true for all $t \in \text{supp } \pi$

and ω by the positive cross-derivative for the copulas.³⁷

Thus $\hat{C}^{MM}(u, v) \leq \hat{C}^{IM}(u, v) \quad \forall u, v \in [0, 1]$ by the equivalence of concordance order and positive quadrant dependence. $\square$

PROOF OF PROPOSITION 2. First consider linear utility with $\omega_i = \pi_i^{IM} + \tilde{r}_i^{IM}$. A direct application of Theorem 3.3 in Navarro and Sarabia (2022) using Assumption 3 gives

$$\omega_i^{IM} = \pi_i^{IM} + r_i^{IM} + \varepsilon_i^{IM} \preceq \pi_i^{MM} + r_i^{MM} + \varepsilon_i^{MM} = \omega_i^{MM} \longrightarrow F_{\omega}^{MM}(p) \leq F_{\omega}^{IM}(p) \quad \forall p. \quad (\text{A36})$$

As the usual stochastic order is preserved under increasing transformations, the stated proposition then follows immediately for any increasing $u(\cdot)$ $\square$

PROOF OF LEMMA 3. Identical in distribution implies $\pi^{MM} \succeq \pi^{IM}$. Thus, each element of the vector $(\pi_i, \tilde{r}_i)$ in mature markets is larger in the usual stochastic order than in immature markets. It follows that the vector $\mathbf{X}^{IM} \preceq \mathbf{X}^{MM}$ in the usual *multivariate* stochastic ordering (Shaked and Shanthikumar 2007, Theorem 6.B.14).

This order is preserved for any increasing function $u : \mathcal{R}^2 \rightarrow \mathcal{R}$, then $u(\mathbf{X}^{IM}) \preceq u(\mathbf{X}^{MM})$. Thus, $F_{\omega}^{IM}(p) \geq F_{\omega}^{MM}(p) \quad \forall p$ and the stated lemma follows. This result extends to any multi-dimensional vector (for example r and ε) under corresponding assumptions on their marginals. $\square$

PROOF OF LEMMA 4. Analogously to the proof of Proposition 1

$$F_{\pi|\omega}^{IM}(\pi|\omega) \succeq F_{\pi|\omega}^{MM}(\pi|\omega) \longrightarrow MC^{IM}(p) \geq MC^{MM}(p) \quad \forall p. \quad (\text{A37})$$

As the unconditional first order stochastic dominance of private valuations in mature and immature markets implies $F_{\omega}^{MM}(\omega) \leq F_{\omega}^{IM}(\omega)$, (A37) holds if and only if the copulas

³⁷This follows from the definition of the conditional density above.

satisfy

$$\widehat{C}_2^{MM}(u, a) \leq \widehat{C}_2^{IM}(u, b) \quad \forall a \geq b \text{ and } \forall u. \quad (\text{A38})$$

This is trivially satisfied under identical concave copulas ($\widehat{C} = \widehat{C}^{IM} = \widehat{C}^{MM}$). However, it cannot be satisfied for different copulas that are both concave in their second argument (Proposition 1 condition(i)). $\square$

PROOF OF PROPOSITION 4. Recall the following necessary and sufficient condition for the comparison of average costs (See the proof of Proposition 1):

$$F_{\pi|\omega \geq p}(\pi|\omega \geq v_1) \leq F_{\pi|\omega \geq p}(\pi|\omega \geq v_2) \quad \forall v_1 \geq v_2 \text{ and } \forall \pi \quad (\text{A39})$$

$$\frac{u - \widehat{C}(u, F_\omega(v_1))}{1 - F_\omega(v_1)} \leq \frac{u - \widehat{C}(u, F_\omega(v_2))}{1 - F_\omega(v_2)} \quad \forall v_1 \geq v_2 \text{ and } \forall u \quad (\text{A40})$$

The analogous condition to order average costs between markets is:

$$F_{\pi|\omega \geq p}^{IM}(\pi|\omega \geq v) \leq F_{\pi|\omega \geq p}^{MM}(\pi|\omega \geq v) \quad \forall v, \pi \quad (\text{A41})$$

$$\frac{u - \widehat{C}^{IM}(u, F_\omega^{IM}(v))}{1 - F_\omega^{IM}(v)} \leq \frac{u - \widehat{C}^{MM}(u, F_\omega^{MM}(v))}{1 - F_\omega^{MM}(v)} \quad \forall v, \forall u \quad (\text{A42})$$

As $F_\omega^{MM}(\omega) \leq F_\omega^{IM}(\omega)$, established in Proposition 2, the following is true for all copulas satisfying condition (ii) of Proposition 1 (see the proof above)

$$\frac{u - \widehat{C}^{MM}(u, F_\omega^{IM}(v))}{1 - F_\omega^{IM}(v)} \leq \frac{u - \widehat{C}^{MM}(u, F_\omega^{MM}(v))}{1 - F_\omega^{MM}(v)} \quad \forall v \text{ and } \forall u, \quad (\text{A43})$$

and by the concordance ordering established in Lemma 2 we have

$$\frac{u - \widehat{C}^{IM}(u, F_\omega^{IM}(v))}{1 - F_\omega^{IM}(v)} \leq \frac{u - \widehat{C}^{MM}(u, F_\omega^{IM}(v))}{1 - F_\omega^{IM}(v)} \quad \forall v \text{ and } u \in [0, 1], \quad (\text{A44})$$

which completes the proof. $\square$

PROOF OF PROPOSITION 5. Note that the copula defines *rank* dependence. Consequently, they are invariant under strictly monotone transformations, including the inverse demand transformation considered here. Thus, to order average costs given quantity, q , we have the following condition:

$$\frac{u - \hat{C}^{IM}(u, q)}{1 - q} \leq \frac{u - \hat{C}^{MM}(u, q)}{1 - q} \quad \forall q \in [0, 1) \text{ and } u \in [0, 1], \quad (\text{A45})$$

which simplifies to

$$\hat{C}^{IM}(u, q) \geq \hat{C}^{MM}(u, q) \quad \forall q \in [0, 1) \text{ and } u \in [0, 1]. \quad (\text{A46})$$

This only depends on the change in sorting ($\hat{C}^{IM} \neq \hat{C}^{MM}$) and holds with equality under no re-sorting ($\hat{C} = \hat{C}^{IM} = \hat{C}^{MM}$). Equation (A38) gives an identical condition for marginal costs. The stated Proposition then follows from the concordance ordering established in Lemma 2. $\square$

Appendix C. Proofs for Section 4

Proof of the first part. Let $-C < 0$ denote the loss when both firms enter and $\Pi_M^{IM} < 0$ denote the monopolist profit in immature markets, respectively. Equivalently Π_M^{MM} denotes the monopolist profit and $\Pi^{MM} = 0$ denotes the (competitive) profit in mature markets.

		enter	exit			enter	exit
enter		(0, 0)	$(\Pi_M^{MM}, 0)$		enter	(-C, -C)	$(\Pi_M^{IM}, 0)$
exit		$(0, \Pi_M^{MM})$	(0, 0)		exit	$(0, \Pi_M^{IM})$	(0, 0)
		(a) Mature Market				(b) Immature Market	

TABLE A1. Second Period Payoff Matrix

Payoffs in the second subgame when at least one firm entered in the first period are given in Panel A of Table A1.

In a mature market that does not fully unravel $\Pi_M^{MM} > 0$ and entering is a profitable deviation for both firms if neither enter, thus this cannot be a nash equilibrium of the subgame. There is no profitable deviation from the symmetric strategy profile (enter, enter) as each firm also earns zero under unilateral deviation.

Next, consider the second subgame when neither firm entered in the first period (Panel B). When $\Pi_M^{IM} < 0$, exiting is a strictly dominant strategy and there is a unique equilibrium.

Taking these contingent strategy profiles as given, the normal form representation of the first subgame is in Table A2. As the market is immature and firms earn zero equilibrium profits in mature markets, these payoffs are identical to the immature market second period subgame and exit is strictly dominant.

Therefore, the following symmetric strategy profile is an SPE:

- First period: Play *exit*

	enter	exit
enter	$(-C + 0, -C + 0)$	$(\Pi_M^{IM} + 0, 0 + 0)$
exit	$(0 + 0, \Pi_M^{IM} + 0)$	$(0 + 0, 0 + 0)$

TABLE A2. First Period

- Second period: Play *exit* if no firm has entered, otherwise *enter*

and in equilibrium no firm enters in either period.

Proof of second part. When the market is mature in the second period, the SPE has firms playing strictly dominant strategies. Therefore, only refinements for the immature second period subgame need to be considered.

As this is a two player, two action game, the set of trembling hand perfect equilibria is equivalent to the set of equilibria in which no player plays a weakly dominated strategy. The equilibria considered above (exit, exit) satisfies this. However, the asymmetric strategy profiles in the second round, either (entry, exit) or (exit, entry), do not because entering is weakly dominated by exiting.

This leaves a unique equilibria surviving the trembling-hand refinement in this subgame. The same argument can be applied in the same fashion to the first period delivering a unique equilibria of the extensive form game.